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Configurations of lines in surfaces

Mikkel Gjestrud

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Mikkel Gjestrud

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Supervisor:
Jørgen Vold Rennemo

Abstract

This thesis studies configurations of lines contained in algebraic surfaces in projective three-space $\mathbb{P}^3$. Several new configurations are defined and studied, and the existence of inclusions of these configurations in surfaces of varying degree is investigated, using both theoretical and computational methods.

Sammendrag

Denne oppgaven studerer konfigurasjoner av linjer som ligger på algebraiske flater i det projektive rommet $\mathbb{P}^3$. Flere nye konfigurasjoner blir definert og studert, og eksistensen av inklusjoner av disse konfigurasjonene i flater av ulik grad blir undersøkt, ved bruk av både teoretiske og numeriske resultater.

Declaration of AI assistance

In this scientific work, generative artificial intelligence (AI) has been used. All data and personal information have been processed in accordance with the University of Oslo's regulations, and I, the author of the document, take full responsibility for its content, claims, and references. An overview of the use of generative AI is provided below.

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x

Chapter 1

Introduction

In 1849, Arthur Cayley communicates to George Salmon that a general cubic surface contains a finite number of lines, and later Salmon responded saying it is exactly 27. Later this same year, Salmon proves that in fact *any* non-singular surface contains exactly 27 lines. Dolgachev claims this can be considered the beginning of modern algebraic geometry ([Dol04]). What can be said for sure is that the study of cubic surfaces still captures the interest of algebraic geometers today. Dolgachev start the article with a quote from Archibald Henderson in his book [Hen11]: "While it is doubteless true that the classification of cubic surfaces is complete, the number of papers dealing with these surfaces which continue to appear from year to year furnish abundant proof of the fact that they still possess much of the same fascination as they did in the days of their discovery of the twenty-seven lines upon the cubic surface." And to this day, research papers on cubic surfaces are still being published. In fact, this thesis includes a novel result on cubic surfaces, namely that one of the configurations explored is generally included in a unique cubic surface. So it is not likely that this fascination will end any time soon.

More than just focusing on the cubic surfaces themselves, the study of lines on surfaces has expanded far beyond just cubic surfaces. The study of lines on hypersurfaces in general, and more generally the study of k-planes on hypersurfaces, is a rich area of research in algebraic geometry. The classical study of lines on hypersurfaces sits at a crossroads of enumerative geometry, birational geometry and modern moduli theory. Passing from the classical era to the modern language, one packages these lines into the *Fano scheme* of a hypersurface. For a smooth cubic surface in $\mathbb{P}^3$ the Fano scheme of lines is a reduced zero-dimensional scheme of length 27; for higher degree hypersurfaces it can have positive dimension and carries vector bundle data coming from restriction of the universal bundles on the Grassmannian.

In this thesis, we will study lines on surfaces in $\mathbb{P}^3$, and in particular we will study certain configurations of lines on these surfaces. These configurations are the following three:

- An *m-cross* is a configuration of m distinct lines all passing through a single point.
- An *m-gon* of lines is an ordered collection of distinct lines $(L_1, \dots, L_m)$ with $L_i \cap L_{i+1} \neq \emptyset$ (indices modulo m) and with no other intersections among non-consecutive pairs. Thus, they form a cycle in the geometric sense.
- An *m-chain* is a path $(L_1, \dots, L_m)$ of distinct lines with $L_i \cap L_{i+1} \neq \emptyset$ for $1 \leq i < m$. Unlike the m-gon it does not wrap around.

Precise definitions appear later.

We study when such configurations can be found on surfaces of a given degree, and in some cases we prove uniqueness results for surfaces containing such configurations. A novel result proved here is that a general 6-gon of lines are contained in unique singular surfaces of 3.

1.1 Outline

The aim of this thesis is to study the m-crosses, m-gons, and m-chains. We also provide explicit computational examples illustrating the phenomena discussed. We describe the theory of lines and higher dimensional subspaces of projective varieties to aid in our goals. The paper is organized as follows:

Later in this chapter we collect some necessary results and notation.

In chapter 2 we introduce the Grassmannian, and give it the structure of a projective variety. We introduce its universal bundles, and we introduce the universal Fano scheme of lines on hypersurfaces.

Chapter 3 introduces intersection theory on divisors, and continues with the Chow ring. We then develop theory on Chern classes of vector bundles. Finally, we prove that a smooth cubic surface contains exactly 27 lines.

In chapter 4 we introduce m-crosses, m-gons and m-chains of lines. We study existence and classification results. We prove the novel result that a general 6-gon is only included in unique singular surfaces of degree 3.

The final chapter 5 summarizes the results and lists open questions and possible directions for further work, as well as conjectures based on computational results.

Appendix A describes the computational methods and implementations used, and presents concrete examples and data.

1.2 Notation

We will use the following notation throughout the thesis:

- We work over the field of complex numbers $\mathbb{C}$ unless otherwise stated.
- $K(X)$ denotes the function field of a variety X.
- $K(X)^{\times}$ denotes the multiplicative group of non-zero elements in $K(X)$.
- We denote $\dim_{\mathbb{C}} H^i(-,-)$ by $h^i(-,-)$.

1.3 Important results

We will now present some important results that will be used later in the thesis.

First, a *linear system* on a scheme X is a pair $\mathcal{D} = (\mathcal{L}, V)$, where $\mathcal{L}$ is a line bundle on X and $V \subseteq H^0(X, \mathcal{L})$ a vector space of sections. We have the following theorem:

Theorem 1.3.1 (Bertini's theorem). *If $\mathcal{D}$ is a finite dimensional linear system on a variety X in characteristic 0, then the general member of $\mathcal{D}$ is smooth away from the base locus of $\mathcal{D}$ and the singular locus of X.*

We refer to [Har77, Corollary III.10.9] for a proof.

Theorem 1.3.2 (Bézout's theorem). *Let $X, Y \subseteq \mathbb{P}^n$ be two closed subvarieties having complementary dimension, i.e.* $\dim(X)+\dim(Y) = n$. *If X and Y intersect transversely, then they will intersect in exactly* $\deg(X)\cdot\deg(Y)$ *points. Furthermore, if the intersection is finite, then the number of intersection points (counted with multiplicity) is at most* $\deg(X) \cdot \deg(Y)$.

We refer to [EH16, Corollary 2.4] for a proof.

Theorem 1.3.3 ([Vak25] Corollary 12.4.2). *Suppose $\pi\colon X \to Y$ is a morphism of irreducible k-varieties, with* $\dim X = m$ *and* $\dim Y = n$. *Then there exists a nonempty open subset $U \subseteq Y$ such that for all $q \in U$, the fiber over q has pure dimension $m - n$, or is empty.*

Theorem 1.3.4 ([Vak25] Corollary 12.2.1). *If X is an irreducible k-variety, then* $\dim X = \mathrm{trdeg} K(X)/k$.

Theorem 1.3.5 (Dimension formula for general fibers). *Let k be a field and let $f\colon X \to Y$ be a dominant morphism of irreducible varieties over k. Let η denote the generic point of Y and write $X_\eta = X \times_Y \operatorname{Spec} k(Y)$ for the generic fiber. Then*

$$\dim X \ = \ \dim Y \ + \ \dim X_\eta.$$

The proof comes from applying the two theorems above.

Lemma 1.3.6. *Let $\pi\colon X \to Y$ be a proper morphism to an irreducible variety, and all the fibers are nonempty and irreducible of the same dimension. Then X is irreducible.*

This can be found in [EO25, Exercise 14.4.21].

Theorem 1.3.7 (Upper semicontinuity of fiber dimension). *Let k be a field, and suppose $\pi\colon X \to Y$ is a morphism of finite type k-schemes. If π is proper, then the dimension of the fiber of π over $q \in Y$ is an upper semicontinuous function in q (i.e., on Y). That is, for each integer $n \geq 0$, the set*

$$\{y \in Y \mid \dim f^{-1}(y) \geq n\}$$

is closed in Y.

This can be found in [Vak25, Theorem 12.4.3].

Chapter 2

Grassmannians and the space of lines in $\mathbb{P}^3$

In this chapter we will introduce Grassmann varieties, or Grassmannians, and focus on the space of lines in $\mathbb{P}^3$, which is denoted $\mathbb{G}(1,3)$. We will also describe the Grassmannians as projective varieties by introducing its affine cover, and a way to embed the Grassmannian into a projective space. For a more detailed introduction to Grassmannians, we refer to Chapter 3 and 4 in [EH16], or [EO25, Section A.8]. We closely follow the exposition in [EO25].

2.1 Definition and basic properties

A Grassmannian is a projective variety whose closed points correspond to linear subspaces of a certain dimension in a vector space. As a set, $\mathrm{Gr}(k, V)$ is the set of all k-dimensional linear subspaces of the n-dimensional vector space V. In our case, V will be a vector space over an algebraically closed field $\mathbb{C}$.

Definition 2.1.1. We define the *Grassmannian* $\mathrm{Gr}(k, V)$ of k-dimensional linear subspaces of V as the set

$$\mathrm{Gr}(k,n)(\mathbb{C}) = \{[L] \mid L \subseteq \mathbb{C}^n \text{ is a linear space of dimension } k\}.$$

A k-dimensional linear subspace of an n-dimensional vector space V can also be thought of as a $(k-1)$-dimensional linear subspace of $\mathbb{P}V \cong \mathbb{P}^{n-1}$. Due to this, we will often write $\mathrm{Gr}(k, V)$ as $\mathbb{G}(k-1, \mathbb{P}V)$. When there is no need to specify the vector space V, only its dimension n, we will write $\mathrm{Gr}(k,n) = \mathbb{G}(k-1, n-1)$. In this thesis we will most often write $\mathbb{G}(k-1, n-1)$, as it will be more natural when working with linear subspaces of projective space. For example, $\mathbb{G}(1,3)$ parametrizes lines in $\mathbb{P}^3$.

The Grassmannian is an important and very useful structure in algebraic geometry. Naturally, when asking questions regarding lines, planes, or other linear subspaces of projective spaces, many answers include the geometry of this variety of the linear subspaces themselves.

2.2 Covering by affine spaces

To describe the Grassmannian as a projective variety, we will describe an affine cover of the Grassmannian. [EO25, Section A.8] goes into detail about this construction, and shows that the cover glues to a scheme. We will go over the overall idea, but exclude the finer details.

We represent a linear subspace $L \subseteq \mathbb{C}^n$ as the kernel of a surjective linear map $\mathbb{C}^n \to \mathbb{C}^{n-k}$. This map can be represented by an $(n-k) \times n$ matrix A of rank $(n-k)$. We have $L = \mathrm{Ker}(A) \subseteq \mathbb{C}^n$, where

$$A = \begin{pmatrix} a_{1,1} & \cdots & a_{1,n} \\ \vdots & \ddots & \vdots \\ a_{(n-k),1} & \cdots & a_{(n-k),n} \end{pmatrix}.$$

Note that the matrix A is not unique for a given linear subspace L, since multiplying A on the left by an invertible $(n-k) \times (n-k)$ matrix $G \in \mathrm{GL}_{n-k}(\mathbb{C})$ gives another matrix GA with the same kernel as A. For this reason, we consider the equivalence relation on the set of $(n-k) \times n$ matrices of rank $(n-k)$ given by $A \sim A'$ if there exists an invertible $(n-k) \times (n-k)$ matrix G such that $A' = GA$. The Grassmannian $\mathrm{Gr}(k, n)$ can then be identified with the set of equivalence classes of $(n-k) \times n$ matrices of rank $(n-k)$ under this equivalence relation. We notice that if L is represented by a matrix A, we can find a more canonical example by instead regarding A in its reduced echelon form. In this way, we represent L by a matrix A with some $(n-k) \times (n-k)$ identity matrix as a submatrix. At the same time, a matrix A determines a subspace $L \subseteq \mathbb{C}^n$, and two matrices in reduced echelon form represent the same subspace if and only if they are equal. Note that a matrix in reduced echelon form with an $(n-k) \times (n-k)$ identity matrix as a submatrix is parametrized by an affine space of dimension $(n-k)k$.

Example 2.2.1. The matrices

$$\begin{pmatrix} 1 & 0 & 1 & 1 \\ 1 & 1 & 0 & 1 \end{pmatrix} \sim \begin{pmatrix} 2 & 1 & 1 & 2 \\ 1 & 1 & 0 & 1 \end{pmatrix}$$

since they represent the same 2-dimensional subspace of $\mathbb{C}$, and is in the same class as the matrix their reduced echelon form

$$\begin{pmatrix} 1 & 0 & 1 & 1 \\ 0 & 1 & -1 & 0 \end{pmatrix}$$

▲

For these reasons, we consider the set

$$(\mathbb{A}^{(n-k)n}(\mathbb{C}) - Z)/\mathrm{GL}_{n-k}(\mathbb{C})$$

where $\mathbb{A}^{(n-k)n}(\mathbb{C})$ parametrizes the set of all $(n-k) \times n$ matrices with entries in $\mathbb{C}$, and Z is the closed subset of $(n-k) \times n$ matrices of rank less than $(n-k)$. The $\mathbb{C}$-points of $\mathrm{Gr}(k, n)$ is bijective to the classes in this set. We now define an open cover of $\mathrm{Gr}(k, n)$.

Let $I \subseteq \{1, \ldots, n\}$ be a subset of size $(n-k)$. For each I let $U_I \subseteq \mathbb{A}^{(n-k)n}(\mathbb{C})$ denote the closed subscheme such that the columns given by I form a $(n-k) \times (n-k)$ submatrix which is the identity matrix. Each U_I is isomorphic to the affine space $\mathbb{A}^{(n-k)k}$. The collection of all such U_I for all subsets I of size $(n-k)$ gives an open cover of $\mathrm{Gr}(k, n)$. Note that there are $\binom{n}{n-k} = \binom{n}{k}$ such subsets, one for each I, which glues together to the Grassmannian the structure of a projective scheme.

Example 2.2.2. For $n = 4, k = 2$, there are 6 affine spaces $\mathbb{A}^4$. They are:

$$U_{12} = \begin{pmatrix} 1 & 0 & a_{1,3} & a_{1,4} \\ 0 & 1 & a_{2,3} & a_{2,4} \end{pmatrix} \qquad U_{13} = \begin{pmatrix} 1 & a_{1,2} & 0 & a_{1,4} \\ 0 & a_{2,2} & 1 & a_{2,4} \end{pmatrix}$$
$$U_{14} = \begin{pmatrix} 1 & a_{1,2} & a_{1,3} & 0 \\ 0 & a_{2,2} & a_{2,3} & 1 \end{pmatrix} \qquad U_{23} = \begin{pmatrix} a_{1,1} & 1 & 0 & a_{1,4} \\ a_{2,1} & 0 & 1 & a_{2,4} \end{pmatrix}$$
$$U_{24} = \begin{pmatrix} a_{1,1} & 1 & a_{1,3} & 0 \\ a_{2,1} & 0 & a_{2,3} & 1 \end{pmatrix} \qquad U_{34} = \begin{pmatrix} a_{1,1} & a_{1,2} & 1 & 0 \\ a_{2,1} & a_{2,2} & 0 & 1 \end{pmatrix}$$

▲

One can also show that this construction is possible over all rings by working over the base ring $\mathbb{Z}$. We will not go into details here, but refer again to [EO25, Section A.8].

Proposition 2.2.3. *For a field $\mathbb{C}$, the Grassmannian $\mathrm{Gr}_{\mathbb{C}}(k, n)$ is a regular projective variety of dimension $k(n - k)$.*

Proof. By construction, the Grassmannian has an affine covering of affine spaces $\mathbb{A}^{(n-k)k}$, so it is regular of dimension $k(n - k)$. □

Example 2.2.4. The Grassmannian $\mathbb{G}(1, 3)$ parametrizing lines in $\mathbb{P}^3$ is a regular projective variety of dimension 4. ▲

2.3 The Plücker embedding

Again we follow [EO25] in describing the Plücker embedding, and keep the exposition brief. We look at $\mathbb{P}^N_{\mathbb{C}}$ for $N = \binom{n}{k} - 1$. Then there is a closed embedding

$$\rho\colon \mathrm{Gr}(k, n) \to \mathbb{P}^{\binom{n}{k}-1},$$

called the *Plücker embedding.* Locally, consider the morphism

$$\rho_I\colon U_I \to \mathbb{P}^{\binom{n}{k}-1}$$

defined by setting for each ring R, and $A \in U_I(R)$

$$\rho_I(A) = [\det(A_J)]_{J \subseteq \{1,\dots,n\}, |J|=k}.$$

That is, ρ_I sends a matrix to all of its $(n - k) \times (n - k)$ minors. These morphisms ρ_I glue together to give the Plücker embedding ρ.

Lemma 2.3.1. *For each I, the morphism ρ_I is a locally closed embedding.*

The proof of this lemma can be found in [EO25, Lemma A.24]. This means that $\mathrm{Gr}(k, n)$ embeds as a closed subscheme of $\mathbb{P}^{\binom{n}{k}-1}$. The image of this embedding is described by the vanishing of certain homogeneous polynomials called the Plücker relations.

2.4 Universal sub and quotient bundles

Later, we will need the notion of the universal subbundle over the Grassmannian. Here we follow the exposition in [EH16, Section 3.2.3]. Let V be an n-dimensional vector space over a field $\mathbb{C}$. Consider the Grassmannian $\mathrm{Gr}(k, V)$ of k-dimensional linear subspaces of V. Let $\mathcal{V} = \mathrm{Gr}(k, V) \times V$ be the trivial vector bundle over $\mathrm{Gr}(k, V)$ with fiber V.

Definition 2.4.1. The *universal subbundle* $\mathcal{S}$ over $\mathrm{Gr}(k, V)$ is the vector bundle whose fiber over a point $[L] \in \mathrm{Gr}(k, V)$ is the subspace $L \subseteq V$. That is,

$$S_{[L]} = L \subseteq V = \mathcal{V}_{[L]}.$$

Definition 2.4.2. The quotient $\mathcal{Q} = \mathcal{V}/\mathcal{S}$ is called the *universal quotient bundle* over $\mathrm{Gr}(k, V)$, and can be viewed as the vector bundle whose fiber over a point $[L] \in \mathrm{Gr}(k, V)$ is the quotient space V/L.

Proposition 2.4.3 ([EH16], Proposition 3.3). *The subset $\mathcal{S}$ of $\mathcal{V}$ whose fiber over a point $[L] \in \mathrm{Gr}(k, V)$ is the subspace $L \subseteq V$ is a vector bundle over $\mathrm{Gr}(k, V)$.*

Later we will use the universal subbundle to prove that a smooth cubic surface contains exactly 27 lines.

2.5 Incidence correspondence and the universal Fano scheme

In this section, we aim to introduce the concept of incidence correspondences (or an incidence variety), which will be a tool we will use several times later in this thesis. We follow the exposition in [EH16, Section 6.1]. We introduce the universal Fano scheme, which is an example of an incidence variety. Generally, an incidence correspondence is a set of pairs (x, y) such that x lies in y, where x and y are elements of X and Y respectively, for some varieties X and Y parametrizing geometric objects. This can be visualized in the following diagram, with projection maps π_1 and π_2:

$$\begin{tikzcd}[column sep=small] & \mathcal{I} = \{(x, y) \in X \times Y \mid x \text{ lies in } y\} \arrow[dl, "\pi_1"'] \arrow[dr, "\pi_2"] & \\ X & & Y \end{tikzcd} \tag{2.1}$$

Now we define an incidence correspondence of k-planes $L \in \mathbb{G}(k, n)$ contained in degree d surfaces in $\mathbb{P}^n$.

Definition 2.5.1. Let $\mathrm{N} = \binom{n+d}{d} - 1$, so that $\mathbb{P}^N$ parametrizes all degree d hypersurfaces in $\mathbb{P}^n$. The *universal Fano scheme* is defined as

$$\Phi(n, d, k) = \{(X, L) \in \mathbb{P}^N \times \mathbb{G}(k, n) \mid L \subseteq X\}$$

Remark 2.5.2. The *Fano scheme* $F_k(X) \subseteq \mathbb{G}(k, n)$ of a variety $X \subseteq \mathbb{P}^n$ parametrizes k-planes on X. The universal Fano scheme parametrizes k-planes for all varieties of a certain degree in $\mathbb{P}^n$. By taking the fiber over a point $[X] \in \mathbb{P}^N$, that being the class of the hypersurface $X \subseteq \mathbb{P}^n$, we get the Fano scheme $F_k(X)$ for the specified variety. In this sense, taking the fiber over a "general" X results in a "general" Fano scheme $F_k(X)$. Studying the universal Fano scheme can then tell us what we can "expect" when examining a specific Fano scheme. It is therefore a very useful tool in telling us about a general member of the varieties.

Lemma 2.5.3. *$\Phi \subset \mathbb{P}^N \times \mathbb{G}(k,n)$ is a closed subset of $\mathbb{P}^N \times \mathbb{G}(k,n)$.*

This is shown in [Har92, Chapter 6].

Proposition 2.5.4 ([EH16], Proposition 6.1). *Let $N = \binom{n+d}{d} - 1$. The universal Fano scheme $\Phi = \Phi(n,d,k) \subseteq \mathbb{P}^N \times \mathbb{G}(k,n)$ is a smooth irreducible variety of dimension*

$$\dim \Phi(n,d,k) = N + (k+1)(n-k) - \binom{k+d}{k}.$$

We ask when we expect a general smooth surface X of degree d to contain lines, and remember that this question is answered by looking at the geometry of the Fano scheme $F_k(X)$ for a general smooth surface X. More precisely one asks what dimension the Fano scheme has, therefore deciding what dimension the set of lines in our general surface is. As noted in Remark 2.5.2, $F_k(X)$ can be obtained by taking the fiber over a point $[X] \in \mathbb{P}^N$. We therefore ask what dimension this fiber will have. From Theorem 1.3.5 we get that for a general point $[X]$ with non-empty fiber $\pi^{-1}([X])$, $\dim \pi^{-1}([X]) = \dim \Phi - \dim \mathbb{P}^N$. We define this as $\phi(n,d,k) := \dim \Phi - N$. Now we can answer the question of when we expect lines to be contained in our surface by looking at $\phi(n,d,k)$:

Corollary 2.5.5 ([EH16], corollary 6.2). *(a) The dimension of any component of the family of k-planes on any hypersurface of degree d in $\mathbb{P}^n$ is at least*

$$\phi(n,d,k) := (k+1)(n-k) - \binom{k+d}{k}.$$

(b) If $\phi(n,d,k) < 0$, then the general hypersurface of degree d in $\mathbb{P}^n$ contains no k-planes.

(c) If $\phi(n,d,k) \geq 0$, and the general hypersurface of degree d contains any k-planes, then every hypersurface of degree d contains k-planes, and every component of the family of k-planes on a general hypersurface of degree d has dimension exactly $\phi(n,d,k)$.

To bring this back to a familiar context of lines on surfaces in $\mathbb{P}^3$, we set $n = 3$ and $k = 1$. We then get that $\phi(3,d,1) = 3 - d$. We use this in the following corollary:

Corollary 2.5.6. *For a hypersurface $X \subseteq \mathbb{P}^3$ of degree d, we expect to find no lines on X when $d \geq 4$, finitely many lines when $d = 3$, and positive-dimensional families of lines when $d \leq 2$.*

The proof of this can be found in [EO25, Theorem A.45].

Chapter 3

Intersection theory and Chern classes

3.1 Intersection Theory

An important tool in algebraic geometry is intersection theory. While we will only use it explicitly in a few cases in this thesis, it is a foundational concept which is useful for building intuition. We will follow the treatment of intersection theory of divisors as presented in the first chapter of [Bea96].

Let X be a smooth variety. Recall that the Picard group $\mathrm{Pic}(X)$ is the group of isomorphism classes of line bundles $\mathcal{L}$ (i.e. of invertible sheaves) on X, with addition given by the tensor product, $[\mathcal{L}] + [\mathcal{L}'] = [\mathcal{L} \otimes \mathcal{L}']$. With every divisor D on X, we associate a line bundle $\mathcal{O}_X(D)$, defined locally as the set of rational functions f on X such that $\mathrm{Div}(f) + D \geq 0$. If D is effective, $\mathcal{O}_X(D)$ comes with a canonical section $s \in H^0(X, \mathcal{O}_X(D))$, $s \neq 0$ which is unique up to scalar multiplication, such that $\mathrm{Div}(s) = D$. The map $D \mapsto \mathcal{O}_X(D)$ identifies $\mathrm{Pic}(X)$ with the group of linear equivalence classes of divisors on X.

Let Y be another smooth variety, and consider a morphism $f \colon X \to Y$. Then we can define the pullback of an invertible sheaf with respect to f ([EO25, Section 16.5]) to get a homomorphism $f^* \colon \mathrm{Pic}(Y) \to \mathrm{Pic}(X)$. If f is a dominant morphism, we can also define the pullback f^*D of a Cartier divisor D on Y as a Cartier divisor on X such that $f^*(\mathcal{O}_Y(D)) = \mathcal{O}_X(f^*D)$. See [EO25, Section 20.7] for details

Let X, Y be surfaces. If f is generically finite of degree d, then we define the direct image f_*C of an irreducible curve on X by setting

$$f_*C := \begin{cases} 0, & \text{if } f(C) \text{ is a point,} \\ r\Gamma, & \text{if } f(C) = \Gamma \text{ is a curve and the induced map } C \to \Gamma \text{ is finite of degree } r. \end{cases}$$

We define f_*D for all divisors D on X by linearity.

We now define intersection multiplicity.

Definition 3.1.1. Let C, C' be two distinct irreducible curves on a surface S, $x \in C \cap C'$, $\mathcal{O}_x$ the local ring of S at x. If f and g are local equations of C and C' in $\mathcal{O}_x$, then the *intersection multiplicity* of C and C' at x is defined as

$$m_x(C \cap C') = \dim_{\mathbb{C}} \mathcal{O}_x/(f, g).$$

Definition 3.1.2. If C, C' are two distinct irreducible curves on a surface S, then the *intersection number* of C and C' is defined as

$$C.C' = \sum_{x \in C \cap C'} m_x(C \cap C').$$

Now we want to show how to compute intersection numbers using line bundles. First, for any sheaf $\mathcal{L}$ on S, let $\chi(\mathcal{L}) = \sum_i (-1)^i \dim H^i(\mathcal{L})$ be the Euler-Poincaré characteristic of $\mathcal{L}$. Then we define the intersection product of two line bundles as follows.

Definition 3.1.3. Let $\mathcal{L}, \mathcal{M}$ be two line bundles on a surface S. Then we define their *intersection product* as

$$\mathcal{L}.\mathcal{M} = \chi(\mathcal{O}_S) - \chi(\mathcal{L}^{-1}) - \chi(\mathcal{M}^{-1}) + \chi(\mathcal{L}^{-1} \otimes \mathcal{M}^{-1}).$$

Theorem 3.1.4. *Let C, C' be two distinct irreducible curves on a surface S. Then*

$$\mathcal{O}_S(C).\mathcal{O}_S(C') = C.C'.$$

Proof. See [Bea96, Theorem I.4] □

In other words, we can compute the intersection number of two curves by computing the intersection number of their associated line bundles. If D, D' are two divisors on S we write $D.D'$ for $\mathcal{O}_S(D).\mathcal{O}_S(D')$. Then we have that we can calculate this intersection number by replacing D or D' by linearly equivalent divisors. This allows us to for example compute the self-intersection number of a divisor, denoted $D.D = D^2$. Next we give some applications of this, which also serve as instructive examples of working with intersection theory.

Example 3.1.5 (Example I.9 (a) in [Bea96]). Let $S = \mathbb{P}^2$. We know that $\mathrm{Pic}(S) \cong \mathbb{Z}$. Let C, C' be curves of degree d and d' respectively. Since $C \equiv dL$ and $C' \equiv d'L'$ for distinct lines L, L', we get Bezout's theorem:

$$C.C' = dL.d'L' = dd'(L.L') = dd'.$$

Here L and L' are lines, so $\mathcal{O}(L) \cong \mathcal{O}(L') \cong \mathcal{O}_{\mathbb{P}^2}(1)$. Using the intersection product of line bundles,

$$L.L' = \mathcal{O}_{\mathbb{P}^2}(1).\mathcal{O}_{\mathbb{P}^2}(1) = \chi(\mathcal{O}_{\mathbb{P}^2}) - 2\chi(\mathcal{O}_{\mathbb{P}^2}(-1)) + \chi(\mathcal{O}_{\mathbb{P}^2}(-2)) = 1,$$

since $\chi(\mathcal{O}_{\mathbb{P}^2}) = 1$ and $\chi(\mathcal{O}_{\mathbb{P}^2}(-1)) = \chi(\mathcal{O}_{\mathbb{P}^2}(-2)) = 0$ (all cohomology groups vanish for these negative twists). Thus, $L.L' = 1$. ▲

Example 3.1.6 (Proposition I.8 (a) in [Bea96]). Let C be a smooth curve, and let $f\colon S \to C$ be a surjective morphism from a smooth and projective surface S to C, with F a fiber of f. Then $F^2 = 0$. ▲

Proof. $F = f^*[x]$ for some point $x \in C$. Then there exists a divisor A on C linearly equivalent to x, such that $x \notin A$. Then $F \equiv f^*A$. Since f^*A is a linear combination of fibers of f all distinct from F, we have $F^2 = F.f^*A = 0$. □

Now we consider the case of a blow-up. Let S be a smooth surface, and let $\pi\colon \hat{S} \to S$ be the blow-up of S at a point $p \in S$. Let $E = \pi^{-1}(p)$ be the exceptional divisor, which we recall is isomorphic to $\mathbb{P}^1$. Also recall that for any curve C on S, we can consider the strict transform $\hat{C}$ of C on $\hat{S}$, defined as the closure of $\pi^{-1}(C \setminus \{p\})$ in $\hat{S}$. Then we have the following lemma on the pullback of divisors under the blow-up map.

Lemma 3.1.7 (Lemma II.2 in [Bea96]). *Let C be an irreducible curve on S which passes through p with multiplicity m. Then $\pi^*(C) = \hat{C} + mE$*

Proposition 3.1.8 (Proposition II.3 (b) in [Bea96]). *Let S be a surface, $\pi\colon \hat{S} \to S$ blow up of a point $p \in X$ and $E \subseteq \hat{S}$ the exceptional curve. Let D, D' be two divisors on S. Then $(\pi^* D).(\pi^* D') = D.D'$, $E.(\pi^* D) = 0$, $E^2 = -1$*

Proof. For the first two equalities, we can replace D and D' by linearly equivalent divisors not passing through p, so that their strict transforms do not intersect E, and their intersection numbers can be computed on S. For the last equality, choose a curve C passing through p with multiplicity 1. So its strict transform intersects E transversely at one point, and $\hat{C}.E = 1$. Since $\hat{C} = \pi^* C - E$ by Lemma 3.1.7, and so $(\pi^* C - E).E = 1$. We know from the second equality that $\pi^* C.E = 0$, so $-E^2 = 1$. Then we get $E^2 = -1$. □

Recall that a divisor D on a variety S is *ample* if some positive multiple of D defines an embedding of S into projective space. We now state the Hodge Index Theorem for surfaces.

Theorem 3.1.9 (Hodge Index Theorem). *Let H be an ample divisor on a smooth surface S, and suppose D is a divisor, $D \neq 0$, with $D.H = 0$. Then $D^2 < 0$.*

The proof can be found in [Har77, Theorem V.1.9].

3.2 The Chow ring

A central object in intersection theory is the *Chow ring* of an algebraic variety. The Chow ring encodes information about subvarieties of a given variety, and allows us to compute intersections of these subvarieties in a rigorous way. In this section, we give an introduction to the Chow ring, following [Ful84, Section 1.3] and [EH16, Section 1.2].

Let X be an algebraic scheme. A *k-cycle* on X is a finite formal sum

$$\Sigma_i m_i [V_i]$$

where the V_i are k-dimensional subvarieties of X, and the m_i are integers. The group of k-cycles on X, denoted $Z_k(X)$, is the free Abelian group generated by the k-cycles. To a k-dimensional subvariety V of X corresponds $[V]$ in $Z_k(X)$. The group of all cycles on X is denoted

$$Z(X) = \bigoplus_k Z_k(X),$$

and its elements are called *cycles*.

Remark 3.2.1. $(n-1)$-cycles are Weil divisors.

For any $(k+1)$-dimensional subvariety W of X, and any non-zero rational function $f \in K(W)^\times$, we can define a k-cycle $[\mathrm{div}(f)]$ on X by

$$[\mathrm{div}(f)] = \sum_V \mathrm{ord}_V(f)[V]$$

where the sum is over all codimension-1 subvarieties V of W. Here, $\mathrm{ord}_V(f)$ is the order function on $K(W)^\times$ defined by the local ring $\mathcal{O}_{W,V}$.

A k-cycle α is *rationally equivalent to zero*, written $\alpha \sim 0$, if there are a finite number of $(k+1)$-dimensional subvarieties W_i of X and $f_i \in K(W_i)^\times$ such that

$$\alpha = \sum_i [\mathrm{div}(f_i)].$$

Since $[\text{div}(f^{-1})] = -[\text{div}(f)]$, the cycles rationally equivalent to zero form a subgroup of $\text{Rat}_k(X) \subseteq Z_k(X)$. The quotient group

$$A_k(X) = Z_k(X)/\text{Rat}_k(X)$$

is called the *k-th Chow group* of X, and its elements are called *Chow classes.* We say that two k-cycles are *rationally equivalent* if they lie in the same class in $A_k(X)$.

Definition 3.2.2. The *Chow group* of X is the direct sum

$$A(X) = \bigoplus_k A_k(X).$$

An element $\alpha \in A^k(X)$ is called a *cycle class.* We say that a cycle is *effective* if all its coefficients are non-negative. A cycle class is *effective* if it can be represented by an effective cycle.

Remark 3.2.3. In many cases, it will be natural to study codimension r subvarieties of an n-dimensional variety X, and so we consider the $(n-r)$-th Chow group. We denote this as $A^r(X) = A_{n-r}(X)$. This group is called the *codimension r Chow group* of X.

Theorem 3.2.4 ([EH16] Theorem-definition 1.5). *If X is a smooth quasi-projective variety, then there is a unique product structure on $A(X)$ satisfying the condition:*

If two subvarieties A, B of X are generically transverse, then

$$[A][B] = [A \cap B].$$

This structure makes

$$A(X) = \bigoplus_{c=0}^{\dim X} A^c(X)$$

into an associative, commutative ring, graded by codimension, called the Chow ring *of X.*

3.2.1 Alternative definition of rational equivalence

A more geometric definition of rational equivalence, and therefore perhaps more intuitive to some, is the following.

Definition 3.2.5. Let $\text{Rat}(X) \subseteq Z(X)$ be the subgroup generated by differences of the form

$$\langle \Phi \cap (\{t_0\} \times X) \rangle - \langle \Phi \cap (\{t_1\} \times X) \rangle$$

where $t_0, t_1 \in \mathbb{P}^1$ and Φ is a subvariety of $\mathbb{P}^1 \times X$ not contained in any fiber $\{t\} \times X$. We say that two cycles are *rationally equivalent* if their difference lies in $\text{Rat}(X)$, and we say that two subschemes are rationally equivalent if their associated cycles are rationally equivalent.

[Ful84, Section 1.6] shows that this definition is equivalent to the previous one.

3.2.2 The moving lemma

One of the fundamental results in intersection theory is the *moving lemma*, which allows us to represent Chow classes by cycles that intersect transversely. In fact, this was historically the basis for a proof of Theorem 3.2.4. Generally, the moving lemma extends an idea from divisors, where one can find the intersection product of two divisors, by considering extended divisors using very ample divisors.

Theorem 3.2.6 (The moving lemma). *Let X be a smooth quasi-projective variety.*

(a) For every $\alpha, \beta \in A(X)$ there are generically transverse cycles $A, B \in Z(X)$ with $[A] = \alpha$ and $[B] = \beta$.

(b) The class $[A \cap B]$ is independent of the choice of such cycles A and B.

We refer to [EH16, Theorem 1.6] for a proof.

3.3 Chern classes

Recall that a Cartier divisor is defined through the vanishing loci of sections of line bundles. Chern classes are a way of generalizing this to vector bundles of higher rank.

3.3.1 The first Chern class of a line bundle

We follow [EH16, Section 1.4]. Let $\mathcal{L}$ be a line bundle on a variety X and σ be a rational section. Then on an open affine set U of a covering of X, we can write $\sigma|_U = f_U/g_U$, and define $\mathrm{Div}(\sigma)|_U = \mathrm{Div}(f_U) - \mathrm{Div}(g_U)$. This definition agrees on the intersections of the open affine sets, and thus defines a Cartier divisor on X. Moreover, if τ is another rational section of $\mathcal{L}$, then $\alpha = \sigma/\tau$ is a well-defined rational function, so $\mathrm{Div}(\sigma)$ and $\mathrm{Div}(\tau)$ are rationally equivalent divisors:

$$\mathrm{Div}(\sigma) - \mathrm{Div}(\tau) = \mathrm{Div}(\alpha) \equiv 0 \bmod \mathrm{Rat}(X).$$

Thus, we can define the first Chern class of the line bundle $\mathcal{L}$ as

$$c_1(\mathcal{L}) = [\mathrm{Div}(\sigma)] \in A^1(X),$$

the rational equivalence class $\mathrm{Div}(\sigma)$ in the Chow group $A^1(X)$, for any nonzero rational section σ. This is well-defined, as it does not depend on the choice of rational section σ.

Example 3.3.1. A consequence is that $c_1(\mathcal{O}_{\mathbb{P}^n}(d))$ is the class of any hypersurface of degree d in $\mathbb{P}^n$. ▲

Proposition 3.3.2 ([EH16], Proposition 1.30). *If X is a variety of dimension n, then c_1 is a group homomorphism*

$$c_1 \colon \mathrm{Pic}(X) \to A^1(X).$$

If X is smooth, then c_1 is an isomorphism.

Note that if $Y \subseteq X$ is a divisor on a smooth variety X, then the ideal sheaf of Y is a line bundle denoted $\mathcal{O}_X(-Y)$, and its inverse in the Picard group is denoted $\mathcal{O}_X(Y)$. The inverse of the map c_1 takes $[Y]$ to $\mathcal{O}_X(Y)$.

3.3.2 Chern classes of vector bundles

We now extend the definition of Chern classes to vector bundles of higher rank. Let $\mathcal{E}$ be a vector bundle on an n-dimensional variety X, and let $r = \mathrm{rank}(\mathcal{E})$. We follow the approach of [EH16, Section 5.2], giving an informal characterization rather than a formal definition.

In the case $r = 1$, the Chern class $c_1(\mathcal{L})$ can be viewed as a measure of non-triviality. If $c_1(\mathcal{L}) = 0$, then $\mathcal{L}$ has a nowhere-vanishing section, and $\mathcal{L} \cong \mathcal{O}_X$. We extend this idea to higher rank vector bundles by considering the "degeneracy locus" of collections of sections. Roughly, this is the locus where the sections become linearly dependent in the fibers of $\mathcal{E}$.

More precisely, the bundle $\mathcal{E}$ is said to be trivial if and only if it has r global sections which are everywhere linearly independent. In this case, any set of r general sections will do. We look at the exterior powers of $\mathcal{E}$. The $(r-i)$-th exterior power $\bigwedge^k \mathcal{E}$ is a vector bundle of rank $\binom{r}{r-i}$, and general sections $s_1, \dots, s_{r-i}$ induce sections of $\bigwedge^{r-i} \mathcal{E}$ given by $s_1 \wedge s_2 \wedge \cdots \wedge s_{r-i}$. For $i = 0$, we have a single section $s_1 \wedge s_2 \wedge \cdots \wedge s_r$ of the line bundle $\bigwedge^r \mathcal{E}$, and the zero locus is by definition $c_1(\bigwedge^r \mathcal{E})$. This is a class in $A^1(X)$, and we call it the *first Chern class of* $\mathcal{E}$.

In general, we have that for $(r-i)$ general sections, we can consider the vanishing locus of

$$s_1 \wedge \cdots \wedge s_{r-i} \in \bigwedge^{r-i} \mathcal{E}.$$

This vanishing locus is called the *i-th degeneracy locus* of the sections $s_1, \dots, s_{r-i}$. Given general sections, the i-th degeneracy locus has codimension i in X, and thus defines a class in $A^i(X)$. This class can be computed using Chern classes.

3.3.3 Important results on Chern classes

The *i-th Chern class* $c_i(\mathcal{E})$ of the vector bundle $\mathcal{E}$ is defined as the class of the i-th degeneracy locus of $(r-i)$ general sections of $\mathcal{E}$, i.e. the class of the vanishing locus of $s_1 \wedge \cdots \wedge s_{r-i} \in \bigwedge^{r-i} \mathcal{E}$ for general sections $s_1, \dots, s_{r-i}$.

To characterize the *i-th Chern class* $c_i(\mathcal{E})$ of the vector bundle $\mathcal{E}$, we use the part (b) of the following theorem.

Theorem 3.3.3 ([EH16], Theorem 5.3). *There is a unique way of assigning to each vector bundle $\mathcal{E}$ on a smooth quasi-projective variety X a class $c(\mathcal{E}) = 1 + c_1(\mathcal{E}) + \ldots + c_n(\mathcal{E}) \in A(X)$ in such a way that*

(a) If $\mathcal{L}$ is a line bundle, then $c(\mathcal{L})$ is $1 + c_1(\mathcal{L})$, where $c_1(\mathcal{L}) \in A^1(X)$ is the class of the divisor of zeros minus the divisor of poles of any rational section of $\mathcal{L}$.

(b) If $\tau_0, \dots, \tau_{r-i}$ are global sections of $\mathcal{E}$, and the degeneracy locus D where they are all dependent has codimension i, then $c_i(\mathcal{E}) = [D] \in A^i(X)$.

(c) (Whitney sum formula) If $0 \to \mathcal{E} \to \mathcal{F} \to \mathcal{G} \to 0$ is an exact sequence of vector bundles on X, then $c(\mathcal{E}) = c(\mathcal{F})c(\mathcal{G}) \in A(X)$.

(d) If $\phi\colon Y \to X$ is a morphism of smooth varieties, then

$$\phi^*(c(\mathcal{E})) = c(\phi^*(\mathcal{E})).$$

Theorem 3.3.4 ([EH16], Theorem 5.12, Splitting principle). *Any universal formula involving Chern classes of bundles that is true for bundles that are direct sums of line bundles is true in general.*

Corollary 3.3.5 ([EH16], Corollary 5.5). *If $\mathcal{E}$ is the direct sum of line bundles $\mathcal{L}_i$, then*

$$c(\mathcal{E}) = \prod c(\mathcal{L}_i) = \prod (1 + c_1(\mathcal{L}_i)).$$

The following examples are from [EH16], example 5.13, 5.14, and 5.16.

Example 3.3.6 (Vanishing). If $\mathcal{E}$ is a rank r vector bundle, then $c_i(\mathcal{E}) = 0$ for $i > r$. To see why, apply the splitting principle to write $\mathcal{E} = \bigoplus_{i=1}^{r} \mathcal{L}_i$ for line bundles $\mathcal{L}_i$. Then, since $c(\mathcal{L}_i) = 1 + c_1(\mathcal{L}_i)$, the total Chern class $c(\mathcal{E}) = \prod(1 + c_1(\mathcal{L}_i))$ by Corollary 3.3.5. This is a product of r factors each of degree at most 1, so any Chern class of degree greater than r must be zero. ▲

Example 3.3.7 (Duals). If $\mathcal{E} = \bigoplus \mathcal{L}_i$, Then

$$c(\mathcal{E}^\vee) = \prod c(\mathcal{L}_i^\vee) = \prod (1 - c_1(\mathcal{L}_i))$$

since $c_1(\mathcal{L}^\vee) = -c_1(\mathcal{L})$ when $\mathcal{L}$ is a line bundle. Given this, Whitney's formula gives

$$c_i(\mathcal{E}^\vee) = (-1)^i c_i(\mathcal{E})$$

▲

Example 3.3.8 (Symmetric squares). Suppose $\mathcal{E}$ is a bundle of rank 2. If $\mathcal{E}$ splits as a direct sum $\mathcal{L} \oplus \mathcal{M}$ of line bundles $\mathcal{L}$ and $\mathcal{M}$ with Chern classes $c_1(\mathcal{L}) = \alpha$, and $c_1(\mathcal{M}) = \beta$, then $c(\mathcal{E}) = (1 + \alpha)(1 + \beta)$ by Whitney's formula. Then we get

$$c_1(\mathcal{E}) = \alpha + \beta, \quad c_2(\mathcal{E}) = \alpha\beta.$$

Now, $\mathrm{Sym}^2\mathcal{E}$ splits as $\mathcal{L}^2 \oplus (\mathcal{L} \otimes \mathcal{M}) \oplus \mathcal{M}^2$, so again by Whitney's formula,

$$\begin{aligned} c(\mathrm{Sym}^2\mathcal{E}) &= (1 + 2\alpha)(1 + \alpha + \beta)(1 + 2\beta) \\ &= 1 + 2(\alpha + \beta) + (\alpha^2 + 8\alpha\beta + 2\beta^2) + 4\alpha\beta(\alpha + \beta). \end{aligned}$$

Now we consider these as the Chern classes of $\mathcal{E}$, and we get:

$$1 + 2c_1(\mathcal{E}) + (2c_1(\mathcal{E})^2 + 4c_2(\mathcal{E})) + 4c_2(\mathcal{E})c_1(\mathcal{E})$$

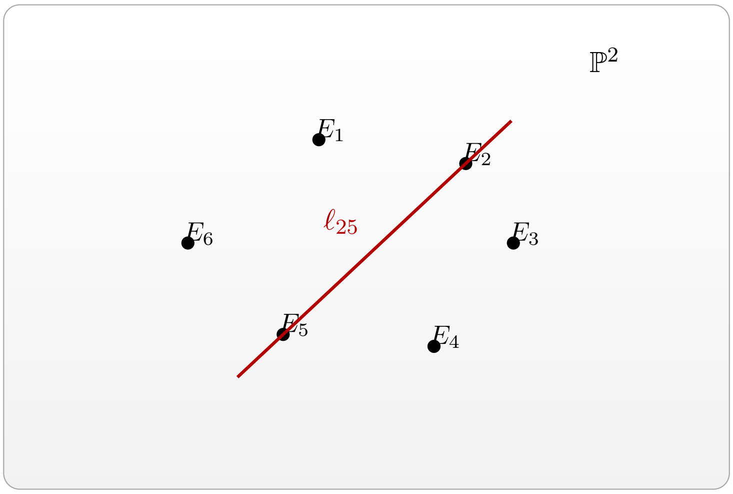
By the splitting principle, this holds in general. ▲

One can use this method to compute Chern classes for any symmetric or exterior power. We will see $\mathrm{Sym}^3\mathcal{E}$ in the next section, which is used to compute the number of lines on a cubic surface. First, we state the following proposition.

Proposition 3.3.9 ([EH16], Proposition 6.4). *Let V be an (n+1)-dimensional vector space, and let $S \subseteq V \otimes \mathcal{O}_\mathcal{G}$ be the tautological rank-(k+1) subbundle on the Grassmannian $\mathcal{G} = \mathcal{G}(k, \mathbb{P}V)$ of k-planes in $\mathbb{P}V \cong \mathbb{P}^n$. A form g of degree d on $\mathbb{P}V$ gives rise to a global section σ_g of $\mathrm{Sym}^d S^\vee$ whose zero locus is the Fano scheme $F_k(X)$, where $X = V(g) \subseteq \mathbb{P}V$ is the hypersurface defined by g. Thus, when $F_k(X)$ has expected codimension $\binom{k+d}{k} = \mathrm{rank}(\mathrm{Sym}^d(S^\vee))$ in $\mathcal{G}$, its class in the Chow ring of $\mathcal{G}$ is given by the top Chern class*

$$[F_k(X)] = c_{\binom{k+d}{k}}(\mathrm{Sym}^d S^\vee) \in A(\mathcal{G}).$$

This is a powerful result, as it allows us to compute the degree of the class of the Fano scheme of k-planes on a general hypersurface using only Chern classes, when the Fano scheme has the expected dimension. We will put this to use in the next section.



Highlighted: strict transform of the line through E_2 and E_5 in $\mathbb{P}^2$.

Figure 3.1: The projective plane $\mathbb{P}^2$ with six marked blow-up points and a strict transform of the line ℓ_{25}.

3.4 27 lines on a cubic surface

From Corollary 2.5.6, we know that a generic smooth cubic surface contains finitely many lines. We will now show that it contains exactly 27 lines. The following is the main theorem of this section.

Theorem 3.4.1. *Each smooth cubic surface in $\mathbb{P}^3$ contains exactly 27 distinct lines.*

We will prove it in two different ways. The first will relate cubic surfaces to blow-ups of $\mathbb{P}^2$ in 6 points, allowing us to picture the lines on smooth cubic surfaces concretely, aiding us later. The second result will use the degree of certain Chern classes to compute the degree of the Fano scheme of lines on a cubic surface, giving us the number of lines directly.

3.4.1 Cubic surfaces as blow-ups of $\mathbb{P}^2$ in 6 points

This proof will rely on theory which is not in the scope of this thesis, but it will allow us to draw a picture to show a representation of the 27 lines themselves, which we will use later.

First, the points $P_1, \dots, P_6$ in $\mathbb{P}^2$ are said to be in *general position* if no three of the points are collinear, and no six lie on a conic.

Proposition 3.4.2 ([Har77], 4.7/4.7.3). *Let $X \subset \mathbb{P}^3$ be a smooth cubic surface. Then there exist points $P_1, \dots, P_6 \in \mathbb{P}^2$ in general position such that X is isomorphic to the blow-up of $\mathbb{P}^2$ in these points.*

This is part of a bigger class of surfaces, those being the *Del Pezzo* surfaces. We refer to [Bea96, Chapter 4] for more information on these.

Proposition 3.4.3 ([Har77], Theorem V.4.9). *() Let $X \subset \mathbb{P}^3$ be a smooth cubic surface. Then X contains exactly 27 lines. Each has self-intersection -1, and they are the only irreducible curves with negative self-intersection. They are:*

- *The exceptional curves $E_1, \dots E_6$ (6 of these)*

- *The strict transform of lines in $\mathbb{P}^2$ containing pairs of points P_i, P_j, $i \neq j$, denoted ℓ_{ij} (15 of these).*
- *The strict transform of conics in $\mathbb{P}^2$ not containing P_i denoted C_i (6 of these).*

$\mathbb{P}^2$ blown up in 6 points is represented in Fig. 3.1.

3.4.2 Proof by degree of Chern classes

Recall from Example 2.2.4 that $\mathbb{G}(1,3)$ has dimension 4.

Proposition 3.4.4 ([EH16], Corollary 6.17). *Let $X \subseteq \mathbb{P}^3$ be a smooth cubic surface of degree $d \geq 3$. If $F_1(X) \neq \emptyset$, then $F_1(X)$ is smooth and zero-dimensional. In particular, X contains at most finitely many lines.*

Sketch of proof of Theorem 3.4.1. By Proposition 3.3.9, the class of $F_1(X)$ of lines on a cubic surface X is given by the class of the Chern class of $c_4(\mathrm{Sym}^3(\mathcal{S}^\vee))$ on $\mathbb{G}(1,3)$. [EH16] shows that

$$\mathrm{Sym}^3(\mathcal{S}^\vee) = \mathcal{L}^3 \oplus (\mathcal{L}^2 \otimes \mathcal{M}) \oplus (\mathcal{L} \otimes \mathcal{M}^2) \oplus \mathcal{M}^3.$$

After computing, they find $c_4(\mathrm{Sym}^3(\mathcal{S}^\vee))$ $27\sigma_{2,2}$, where $\sigma_{2,2}$ is the class of a point in $A(\mathbb{G}(1,3))$. The points in $A(\mathbb{G}(1,3))$ represents a line in $\mathbb{P}^3$. Thus, this gives us that there are 27 lines in a smooth cubic surface.

$\square$

Chapter 4

Configurations of lines

4.1 Surfaces including m-crosses

In this section we will study the problem of finding a smooth hypersurface $X_d \subset \mathbb{P}^3$ of degree d, containing an *m-cross*.

Definition 4.1.1. An m-cross is a union of m distinct lines $l_1, l_2, \ldots, l_m$ in $\mathbb{P}^3$ which intersect at a common point p and lying in a common plane H. We will write $Z_m = \bigcup_{i=1}^{m} l_i$.

We require the lines to lie in a plane because if this is not true, then one can show that the point p gives rise to a tangent space of dimension 3, which cannot be included in a surface in $\mathbb{P}^3$.

It is worth noting the specific question we are asking, which is that given an m-cross, does there exist a smooth hypersurface of degree d including it. We do not ask whether an m-cross is to be expected inside a general hypersurface.

Remark 4.1.2. Note that for automorphisms of $\mathbb{P}^3$, lines are sent to lines. Therefore, we can study the lines $l_1 = (\alpha : \beta : 0 : 0),\ l_2 = (\alpha : 0 : \gamma : 0),\ l_3 = (\alpha : \delta : \lambda_3\delta : 0), \ldots, l_m = (\alpha : \delta : \lambda_m\delta : 0)$ for $\lambda_i \in \mathbb{N}$, where α, β, γ, and λ_i are parameters which vary, by taking the automorphism of $\mathbb{P}^3$ which sends these lines to the given lines, and then study the hypersurface of degree d in $\mathbb{P}^3$ which contains these lines. This automorphism will also take a smooth hypersurface of degree d to a smooth hypersurface of degree d, so showing properties for lines in this form is sufficient for the general case. In this way, we can assume without loss of generality that the plane H containing the lines is given as $H = V(x_3)$, and we notice that a line $l_i = V(H) \cap V(g_i)$ for some $g_i \in \mathbb{C}[x_0, x_1, x_2, x_3]$. The lines given above can then be described as the zero loci of the following polynomials:

$$l_1 = V(x_3, x_2),\ l_2 = V(x_3, x_1),\ l_3 = V(x_3, x_1 - \lambda_3 x_2),\ \ldots,\ l_m = V(x_3, x_1 - \lambda_m x_2). \quad (4.1)$$

We are interested in both the existence and non-existence of degree d hypersurfaces containing m-crosses. We first consider surfaces with degree less than k, but more than 1 (degree 1 provides us a simple choice of surface including an m-cross, namely H).

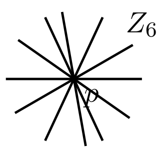


Figure 4.1: An example of a 6-cross in $\mathbb{P}^3$.

Lemma 4.1.3. *If S is a smooth surface in $\mathbb{P}^3$ of degree $d > 1$, it does not contain a plane.*

Proof. Assume for contradiction that $S = V(g)$ contains a plane H. We let $H = V(x_3)$ without loss of generality, by Remark 4.1.2. Then x_3 divides g, and we write $g = x_3h$ for some $h \in \mathbb{C}[x_0, x_1, x_2, x_3]_{d-1}$. We then use the Jacobi criterion [EO25, Proposition 16.16] on x_3h, and get that S is singular on the whole of $V(x_3) \cap V(h)$. This is non-empty by Bezout's theorem (Theorem 1.3.2), giving a contradiction. (Use $d > 1$ to see that $\deg(h) = d - 1 > 0$). □

Lemma 4.1.4. *Given an m-cross, there does not exist any smooth hypersurface of degree $1 < d < m$ such that the lines are contained in the hypersurface.*

Proof. Assume for contradiction that there exists a smooth hypersurface of degree $1 < d < m$ such that the lines are contained in the hypersurface. Then we can fix another line ℓ in H, which does not intersect p, and which does not lie in the hypersurface X_d by Lemma 4.1.3. ℓ will intersect each of the lines at a point p_i, all being distinct since the lines only intersect in p. ℓ will also intersect X_d at the points p_i, thus intersecting the hypersurface at m points. By Bezout's theorem (Theorem 1.3.2), the intersection of a line with a hypersurface of degree d is at most d points, giving a contradiction. □

We now move on to the case of existence, with the following proposition:

Proposition 4.1.5. *For a given m-cross, we can find a smooth hypersurface of degree m such that the lines are contained in the hypersurface.*

Proof. Let $\mathfrak{L}$ be the linear system of hypersurfaces of degree m in $\mathbb{P}^3$ containing the lines $l_1, \ldots, l_m$. Recall that Bertini's theorem, Theorem 1.3.1, tells us that the general member of $\mathfrak{L}$ is smooth outside the base locus of $\mathfrak{L}$. We can therefore narrow our search of singularities to the base locus. The base locus of $\mathfrak{L}$ is the union of the lines $l_1, \ldots, l_m$. To see why, consider a point $x \in \mathbb{P}^3$ which does not lie on one of the lines. We want to show that for each of these points, we can find a surface which excludes the points, but includes the surface. First, if the point is not in the plane H, we can exclude it by the plane itself, so we assume the point lies in H, but not in one of the lines. Then we consider the loci which define the lines, as discussed in Remark 4.1.2, denoted $l_i = V(H) \cap V(g_i)$. Then all the lines $l_1, \ldots l_m$ are included in $V(\Pi_{i=1}^m g_i)$, while x is not.

Since we know exactly the base locus, we want to find a function $f \in \mathbb{C}[x_0, x_1, x_2, x_3]_m$ such that $V(f)$ is smooth on the lines $l_1, \ldots, l_m$. Bertini's theorem then gives us that we can find a hypersurface which is also smooth outside the base locus of $\mathfrak{L}$.

Now we take inspiration from 4.1, and study the function $f = g(x_1, x_2) + h(x_0, x_3)$, where $h = x_0^{m-1}x_3$ and $g = \prod l_i = x_1x_2 \prod_{i=3}^k (x_1 - \lambda_i x_2)$, giving the following polynomial:

$$f(x_0, x_1, x_2, x_3) = x_1x_2 \prod_{i=3}^m (x_1 - \lambda_i x_2) + x_0^{m-1}x_3.$$

We can see that f is a polynomial of degree m, and that $V(f)$ includes the lines $l_1, \ldots, l_m$, since $f(\alpha : \beta : 0 : 0) = f(\alpha : 0 : \gamma : 0) = f(\alpha : \delta : \lambda_i\delta : 0) = 0$. We must then check that f is smooth on the base locus of $\mathfrak{L}$, which is to check that $\bigcap_{i=0}^3 (\partial_i f = 0) = \emptyset$.

The derivatives are given as follows:

$$\partial_0 f = \partial_0 g + \partial_0 h = 0 + (m-1)x_0^{m-2}x_3 = (m-1)x_0^{m-2}x_3,$$

$$\partial_1 f = \partial_1 g + \partial_1 h = x_2 \prod_{i=3}^{m} (x_1 - \lambda_i x_2) + x_1 x_2 \sum_{i=3}^{m} \prod_{\substack{j=3 \\ j \neq i}}^{m} (x_1 - \lambda_j x_2),$$

$$\partial_2 f = \partial_2 g + \partial_2 h = x_1 \prod_{i=3}^{m} (x_1 - \lambda_i x_2) + x_1 x_2 \sum_{i=3}^{m} (-\lambda_i) \cdot \prod_{\substack{j=3 \\ j \neq i}}^{m} (x_1 - \lambda_j x_2),$$

$$\partial_3 f = \partial_3 g + \partial_3 h = 0 + x_0^{m-1} = x_0^{m-1}.$$

We then find that on the base locus of $\mathfrak{L}$, the following hold:

- $\partial_0 f = 0$ on the entire base locus, since it contains the factor x_3, which is 0 on the entire plane.
- $\partial_1 f = 0$ on the line $l_0 = (\alpha : \beta : 0 : 0)$, since $\partial_1 f$ contains the factor x_2, and is non-zero elsewhere on the base locus.
- $\partial_2 f = 0$ on the line $l_1 = (\alpha : 0 : \gamma : 0)$, since $\partial_2 f$ contains the factor x_1, and is non-zero elsewhere on the base locus.
- $\partial_3 f = 0$ if and only if $x_0 = 0$, which is the case for the following points: $(0 : \beta : 0 : 0), (0 : 0 : \gamma : 0), (0 : \delta : \lambda_i \delta : 0)$ for $3 \leq i \leq m$.

We can see that the intersection of all of these zero loci is empty, showing that the function f is smooth on the base locus of $\mathfrak{L}$.

Finally, we create a specific linear system of hypersurfaces of degree m in $\mathbb{P}^3$ containing the lines $l_1, \ldots, l_m$. We take the pencil $\mathcal{D} = \{aV(h) + bV(g) \mid a, b \in \mathbb{C}\}$, where h and g are as above. Then the calculation above shows that the base locus of $\mathcal{D}$ is the m-cross, and that a member $aV(h) + bV(g)$ is smooth on the base locus as long as $a \neq 0$. By Bertini's theorem, the general member of $\mathcal{D}$ is smooth outside the base locus, so we conclude that the general member of $\mathcal{D}$ is smooth.

Thus, we can find a smooth hypersurface of degree k such that the lines are contained in the hypersurface. $\square$

By the results above, we get the following corollary:

Corollary 4.1.6. *Given an m-cross, we can find a smooth hypersurface of degree d such that the lines are contained in the hypersurface if and only if $d \geq m$ or $d = 1$.*

Proof. If we have m lines, we can find a smooth hypersurface of degree $d \geq m+1$ by including arbitrary lines $l_{m+1}, l_{m+2}, \ldots, l_d$ lying in the plane. For the case $d = 1$, we can take the plane H containing the lines. $\square$

4.2 The set of degree d surfaces including m-crosses

In the previous section we studied the existence and non-existence of smooth hypersurfaces of degree d including an m-cross. We will now study the dimension of the set of degree d surfaces including m-crosses. Let Z_m be an m-cross lying in a plane H, which we will assume is the plane $V(x_3)$ by an automorphism of $\mathbb{P}^3$.

Proposition 4.2.1. *Let $A_m = \{S_d \subseteq \mathbb{P}^3 \mid Z_m \subset S_d\} \subseteq \{S_d \subseteq \mathbb{P}^3\}$ be the set of degree d surfaces containing a general Z_m. Then* $\dim A_m = \binom{d-m+2}{2} + \binom{d+2}{3} - 1$.

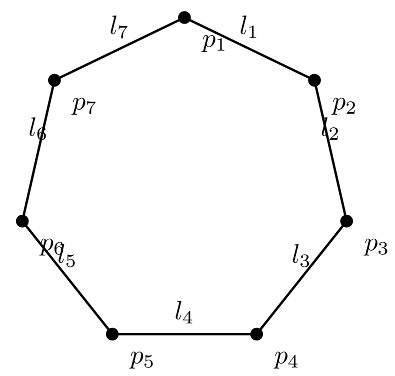


Figure 4.2: 7-gon drawn in a plane; each vertex is an intersection point p_i and the line segments represent the lines l_i. Note that the lines in reality extend infinitely in both directions.

Proof. Let $S_d = V(f)$ for some $f \in \mathbb{C}[x_0, x_1, x_2, x_3]_d$ be a surface which includes Z_m. Since Z_m is included in H, we write f as $f = g(x_0, x_1, x_2) + x_3 h(x_0, x_1, x_2, x_3)$, separating the part of f, $x_3 h(x_0, x_1, x_2, x_3)$, which vanishes on the plane. Saying that $Z_m \subseteq S_d$ is the same as saying $f|_{Z_m} = 0$, which is equivalent to $g(x_0, x_1, x_2)|_{Z_m} = 0$, since $x_3 h(x_0, x_1, x_2, x_3)|_{Z_m}$ already vanishes. This is equivalent to $l_1 l_2 \cdots l_m$ divides $g(x_0, x_1, x_2)$, meaning $g(x_0, x_1, x_2) = l_1 l_2 \ldots l_m \cdot j(x_0, x_1, x_2)$ for some $j(x_0, x_1, x_2) \in \mathbb{C}[x_0, x_1, x_2]_{d-m}$. Thus, we can write f as $f = l_1 l_2 \cdots l_m \cdot j(x_0, x_1, x_2) + x_3 h(x_0, x_1, x_2, x_3)$, where $h(x_0, x_1, x_2, x_3) \in \mathbb{C}[x_0, x_1, x_2, x_3]_{d-1}$.

Since we can choose $j(x_0, x_1, x_2)$ and $h(x_0, x_1, x_2, x_3)$ freely, we get that $\dim A_m = \dim \mathbb{C}[x_0, x_1, x_2]_{d-m} + \dim \mathbb{C}[x_0, x_1, x_2, x_3]_{d-1} - 1 = \binom{d-m+2}{2} + \binom{d+2}{3} - 1$. We subtract 1 to account for the fact that we are working in projective space. $\square$

4.3 Surfaces including m-gons

In this section, we will consider m-gons in $\mathbb{P}^3$ and their relationship with hypersurfaces. Firstm we define what we mean by an m-gon.

Definition 4.3.1. An m-gon is a union of m lines $l_1, l_2, \ldots, l_m$ such that l_i intersects l_{i+1} for $1 \le i < m$, and l_m intersects l_1. We will write $C_m = \bigcup_{i=1}^m l_i$ and denote the intersection points as $p_i = l_i \cap l_{i+1}$ for $1 \le i < m$, and $p_m = l_m \cap l_1$.

Definition 4.3.2. A *regular* m-gon is an m-gon such that $l_i \neq l_j$ for $i \neq j$, and no lines other than the adjacent ones intersect.

The notion of a regular m-gon is important. For example, a non-regular m-gon can be m lines all intersecting at a single point, i.e. an m-cross. Such cases would not hold the same properties as what one would expect from m-gons such as the one drawn in Fig. 4.2.

We will need the following lemma:

Lemma 4.3.3. *The global sections of the structure sheaf of a regular m-gon C_m has dimension $h^0(\mathcal{O}_{C_m}(d)) = md$, for $d \ge 1$.*

Proof. First, consider the short exact sequence:

$$0 \to \mathcal{O}_{C_m}(d) \to \bigoplus_{i=1}^{m} \mathcal{O}_{l_i}(d) \to \bigoplus_{i=1}^{m} \mathcal{O}_{p_i}(d) \to 0.$$

This sequence comes from the fact that sections on C_m can be viewed as collections of sections on each line l_i, which agree at the intersection points p_i, so we note that this argument requires the regularity of C_m. Also note that since twisting $\mathcal{O}_{p_i}$ does not affect it, will write it without the twists going forward. Taking the long exact sequence in cohomology, we get

$$0 \to H^0(\mathcal{O}_{C_m}(d)) \to H^0(\bigoplus_{i=1}^{m} \mathcal{O}_{l_i}(d)) \to H^0(\bigoplus_{i=1}^{m} \mathcal{O}_{p_i}) \to H^1(\mathcal{O}_{C_m}(d)) \to 0,$$

since $H^1(\bigoplus_{i=1}^m \mathcal{O}_{l_i}(d)) = 0$ for all $d \geq 0$. Taking dimensions, we get

$$h^0(\mathcal{O}_{C_m}(d)) = h^0(\bigoplus_{i=1}^{m} \mathcal{O}_{l_i}(d)) + h^1(\mathcal{O}_{C_m}(d)) - h^0(\bigoplus_{i=1}^{m} \mathcal{O}_{p_i}).$$

Since $h^0(\bigoplus_{i=1}^m \mathcal{O}_{p_i}) = m$, and $h^0(\bigoplus_{i=1}^m \mathcal{O}_{l_i}(d)) = m \cdot \binom{d+1}{1}$, this gives us $h^0(\mathcal{O}_{C_m}) = md + h^1(\mathcal{O}_{C_m}(d))$.

Next we will show that $h^1(\mathcal{O}_{C_m}(d)) = 0$. To see this, we show that the map from $H^0(\bigoplus_{i=1}^m \mathcal{O}_{l_i}(d))$ to $H^0(\bigoplus_{i=1}^m \mathcal{O}_{p_i})$ is surjective. As discussed when considering the exact sequence, the map $\bigoplus_{i=1}^m \mathcal{O}_{l_i}(d) \to \bigoplus_{i=1}^m \mathcal{O}_{p_i}$ takes sections $(f_{12}, f_{23}, \ldots, f_{m1})$ to $(f_{12}(p_1) - f_{m1}(p_1), f_{23}(p_2) - f_{12}(p_2), \ldots, f_{m1}(p_m) - f_{(m-1)m}(p_m))$. In this case, we denote f_{ij} for the section on the line from p_i to p_j. Our goal then is to show that we can send these differences to a basis for $\bigoplus_{i=1}^m \mathcal{O}_{p_i}$. We will show that the element $(\alpha, 0, \ldots, 0)$ is in the image of our map, for α non-zero. That is, we are looking for a collection of sections $(f_{12}, f_{23}, \ldots, f_{m1})$ which agree on all points except p_1. Finding sections which agree on points other than p_1 is as simple as letting them be the zero sections, which can all be considered to have arbitrary degrees. Therefore, let $f_{23} = \cdots = f_{(m-1)m} = 0$. Since only one of the sections on p_1 needs to be non-zero, we also let $f_m 1 = 0$. Then we need to find a section f_{12} which evaluates to some α at p_1, and 0 at p_2. Let

$$f_{12} = (\frac{x_0}{\alpha})^d + (\frac{x_1}{\beta})^d + (\frac{x_2}{\gamma})^d - 3(\frac{x_3}{\delta})^d,$$

where $p_2 = (\alpha : \beta : \gamma : \delta) \neq p_1$. This is a degree d section on the line l_1, and evaluates to 0 at p_2, and a nonzero number at p_1.

Thus, we have found sections $(f_{12}, f_{23}, \ldots, f_{m1})$ which map to $(\alpha, 0, \ldots, 0)$. By symmetry, we can find sections mapping to any basis element of $\bigoplus_{i=1}^m \mathcal{O}_{p_i}$, and so the map is surjective and $h^1(\mathcal{O}_{C_m}(d)) = 0$. Therefore, we have that $h^0(\mathcal{O}_{C_m}) = md$.

□

We will add a corollary to this lemma, Corollary 4.4.8, in the next section about m-chains. So be warned that one may need to go back to the lemma above when reading that section. But now we are ready to prove the main proposition of this section.

Proposition 4.3.4. *Let C_m be a regular m-gon in $\mathbb{P}^3$. Then we can find a hypersurface S_d of degree d such that $C_m \subseteq S_d$ when $m < \binom{d+3}{3} \cdot \frac{1}{d}$, or equivalently $m < \frac{(d+3)(d+2)(d+1)}{6d}$.*

Proof. By abuse of notation, we write $\mathcal{O}_{C_m}$ for $i_*\mathcal{O}_{C_m}$, where $i : C_m \to \mathbb{P}^3$ is the inclusion. We are interested in the global sections of twists of the ideal sheaf $\mathcal{I}_{C_m}$ of the m-gon, i.e. $H^0(\mathcal{I}_{C_m}(d))$, to see if the m-gon can be contained in a hypersurface of degree d. To see why, consider the short exact sequence

$$0 \to \mathcal{I}_{C_m} \to \mathcal{O}_{\mathbb{P}^3} \to \mathcal{O}_{C_m} \to 0$$

where $\mathcal{I}_{C_m}$ is the kernel of the restriction map $\mathcal{O}_{\mathbb{P}^3} \to \mathcal{O}_{C_m}$. Since $\mathcal{O}_{\mathbb{P}^3}(d)$ is locally free, we can twist with these to get the short exact sequence

$$0 \to \mathcal{I}_{C_m}(d) \to \mathcal{O}_{\mathbb{P}^3}(d) \to \mathcal{O}_{C_m}(d) \to 0.$$

From this we get the long exact sequence in cohomology:

$$\begin{tikzcd} 0 \arrow[r] & H^0(\mathcal{I}_{C_m}(d)) \arrow[r] & H^0(\mathcal{O}_{\mathbb{P}^3}(d)) \arrow[r] & H^0(\mathcal{O}_{C_m}(d)) \arrow[dll, out=-10, in=170] \\ & H^1(\mathcal{I}_{C_m}(d)) \arrow[r] & H^1(\mathcal{O}_{\mathbb{P}^3}(d)) \arrow[r] & H^1(\mathcal{O}_{C_m}(d)) \arrow[dll, out=-10, in=170] \\ & H^2(\mathcal{I}_{C_m}(d)) \arrow[r] & H^2(\mathcal{O}_{\mathbb{P}^3}(d)) \arrow[r] & \dots \end{tikzcd}$$

By Theorem 20.23 and Corollary 20.26 in [EO25], we know that $H^i(\mathcal{O}_{\mathbb{P}^3}(d)) = 0$ for $i > 0$ and $d \geq (-3)$. Hence, we can focus on the first part of the sequence:

$$0 \to H^0(\mathcal{I}_{C_m}(d)) \to H^0(\mathcal{O}_{\mathbb{P}^3}(d)) \to H^0(\mathcal{O}_{C_m}(d)) \to H^1(\mathcal{I}_{C_m}(d)) \to 0.$$

Now, $H^0(\mathcal{I}_{C_m}(d))$ is precisely the kernel of the restriction map $H^0(\mathcal{O}_{\mathbb{P}^3}(d)) \to H^0(\mathcal{O}_{C_m}(d))$, and so defines the global sections of $\mathcal{O}_{\mathbb{P}^3}(d)$ which vanish on C_m. Therefore, the dimension of this space reveals whether the space of global sections of $\mathcal{O}_{\mathbb{P}^3}(d)$ which vanish on C_m is non-empty or not. Taking dimensions in cohomology is additive, and we get that

$$h^0(\mathcal{I}_{C_m}(d)) = h^0(\mathcal{O}_{\mathbb{P}^3}(d)) + h^1(\mathcal{I}_{C_m}(d)) - h^0(\mathcal{O}_{C_m}(d)).$$

By Corollary 20.26 in [EO25] we know $h^0(\mathcal{O}_{\mathbb{P}^3}(d)) = \binom{d+3}{3}$. By Lemma 4.3.3 we have that $h^0(\mathcal{O}_{C_m}(d)) = md$. So we have that

$$h^0(\mathcal{I}_{C_m}(d)) = \binom{d+3}{3} - md + h^1(\mathcal{I}_{C_m}(d)).$$

$H^1(\mathcal{I}_{C_m}(d)) \geq 0$ measures the failure of the restriction map $H^0(\mathcal{O}_{\mathbb{P}^3}(d)) \to H^0(\mathcal{O}_{C_m}(d))$ to be surjective, i.e. the failure of lifting sections on C_m to global sections. If the map was surjective, we would have that $h^0(\mathcal{I}_{C_m}(d)) = \binom{d+3}{3} - md$. We can at least say that $h^1(\mathcal{I}_{C_m}(d)) \geq 0$, and so

$$h^0(\mathcal{I}_{C_m}(d)) \geq \binom{d+3}{3} - md.$$

Meaning we can guarantee a degree d surface when $\binom{d+3}{3} - md > 0$, or equivalently when $m < \frac{(d+3)(d+2)(d+1)}{6d}$. □

Remark 4.3.5. Note that this argument does not guarantee that the surface is smooth, only that it exists.

To get an exact value for $h^0(\mathcal{I}_{C_m}(d))$, we would need to compute $h^1(\mathcal{I}_{C_m}(d))$. We have no direct answer of what this cohomology is, but the issue will be explored further through examples in the next two sections. We add this corollary to the proposition:

Corollary 4.3.6.

$$h^0(\mathcal{I}_{C_m}(d)) - h^1(\mathcal{I}_{C_m}(d)) = \binom{d+3}{3} - md.$$

Table 4.1: Results from Macaulay2 over $\mathbb{Q}$ testing existence and smoothness of surfaces containing an m-gon. Each row corresponds to one experiment: degree d, number of points used, whether the chosen form was singular, whether the space of degree-d forms in the ideal was zero.

d	m	Singular	Zero
2	4	false	false
2	5	false	true
3	6	false	false
3	7	false	true
4	8	false	false
4	9	false	true
5	11	false	false
5	12	false	true
6	13	false	false
6	14	false	true

4.3.1 Computational results

We have done computational experiments in Macaulay2 to get a better understanding of the inequality in Proposition 4.3.4. The script tests the existence and smoothness of hypersurfaces including an m-gon for various degrees d and number of lines m. The code used for these experiments is included in Section A.1, where we also discuss the details of the code. We note that to run this in Macaulay2, we have run the computation over a finite field $\mathbb{Z}_{32749}$, and over $\mathbb{Q}$, the latter being significantly slower.

We summarize the steps taken in the code for checking if an m-gon is included in some degree d surface:

1. Choose field $\mathbb{C}$, we use either $\mathbb{Q}$ or $\mathbb{Z}_{32749}$, where the latter is faster, but not of characteristic 0.
2. Pick m random points in $\mathbb{P}^3$.
3. Build an m-gon by connecting the points in a cycle.
4. Connect pairs of points to generate the ideal of the line defined by them.
5. Intersect the ideals of the lines to get the ideal of the m-gon.
6. Pick a random element of degree d in the ideal of the m-gon, i.e. pick a random element of $I_Z(d)$. This defines a surface of degree d containing the m-gon.
7. Check if this element is smooth, singular, or equal to 0.

In the table, we also include the number of elements found in $I_Z(d)$, the degree-d part of the ideal of the m-gon. This is to give an idea of how big the space of degree-d forms vanishing on the m-gon is. If this number is 0, then there are no surfaces of degree d containing the m-gon. We calculate this number by first calculating the number of degree-d forms in the polynomial ring, which is the familiar binomial coefficient $\binom{d+3}{3}$.

The results are listed in full in the appendix. The tables are included in Table A.1 and Table A.2, and we have included a smaller table of results over $\mathbb{Q}$ in Table 4.1. They are consistent with the inequality in Proposition 4.3.4, always finding that an m-gon can be included in a surface of degree d when m is lower than the bound, and also always find the surfaces to be smooth. Interestingly, they also imply that we *cannot* find any surface (smooth or singular) containing an m-gon when m equals or exceeds the bound. This suggests that $h^1(\mathcal{I}_{C_m}(d))$ does not exceed $md - \binom{d+3}{3}$ when m gets bigger, though we have not been able to prove this. Our tables include two values of m for each degree d. This is to check both below and above the bound for each degree. If the bound is an integer M, we test $m = M - 1$, and $m = M$. As is shown in the table in the appendix, we have found no cases where we can find a surface including an m-gon when m is equal to or larger than the bound.

Remark 4.3.7. When running the computations in Macaulay2, we have never encountered a case where we have gotten a singular surface one time, and a smooth surface another, even though the surfaces are picked at random. One of the reasons this is perhaps not very surprising, is to consider that being smooth is a Zariski open requirement, meaning the subset of smooth surfaces is open in the set of surfaces. Since we know that this means the subset is dense in the set of surfaces, and so it is not very surprising that the random surface picked is smooth. Still, this adds another reason to not use Macaulay2 as a proof of something, but merely a suggestion that it might be true. One could technically be very unlucky and only pick singular surfaces.

4.3.2 Cohomology of a 4-gon

As mentioned earlier, to get an exact value for $h^0(\mathcal{I}_{C_m})$, we want to compute $h^1(\mathcal{I}_{C_m})$. We will now explore this issue for the case of a 4-gon and looking at a suitable resolution of the ideal sheaf of the 4-gon. First we claim that the intersection points of a 4-gon can, by an automorphism of $\mathbb{P}^3$, be expressed as the points $(1:0:0:0), (0:1:0:0), (0:0:1:0), (0:0:0:1)$. Then the 4-gon can be expressed as the union of the lines $l_1 = V(x_2, x_3), l_2 = V(x_0, x_3), l_3 = V(x_0, x_1), l_4 = V(x_1, x_2)$. Now the ideal of the 4-gon is given by $I_{C_4} = (x_2, x_3) \cap (x_0, x_3) \cap (x_0, x_1) \cap (x_1, x_2) = (x_1x_3, x_0x_2)$. We justify our chose of points in the following lemma.

Lemma 4.3.8. *Let C_4 be a regular 4-gon in $\mathbb{P}^3$. Then there exists an automorphism of $\mathbb{P}^3$ such that the intersection points of C_4 are given by the points* $(1:0:0:0), (0:1:0:0), (0:0:1:0), (0:0:0:1)$.

Proof. Let p_1, p_2, p_3, p_4 be the intersection points of C_4. To be in general position in $\mathbb{P}^3$, we require that no three of the points are collinear, and that the points do not all lie in a plane. We already know that the points do not lie on a plane by considering a regular 4-gon, since if this was included in a plane, then all lines would intersect by Theorem 1.3.2. Let A be the 4×4 matrix whose columns are the homogeneous coordinates of the points. Since we have that the points do not all lie on a plane, we know that the rows are linearly independent, and we can transform the points by multiplying the matrix by A^{-1}. Then $AA^{-1} = I_4$, and we have sent the points to the desired locations. $\square$

To get a value of $h^1(\mathcal{I}_{C_4})$, we will first look at a resolution of the ideal sheaf I_{C_4}. We take inspiration from Hilbert's syzygy theorem, [EO25, Theorem 19.12], which states that we can find a finite free resolution for a finitely generated graded module. We therefore aim to find a resolution. From the discussion above, we have that $I_{C_4} = (x_1x_3, x_0x_2)$ (the calculation was done in Macaulay2). We know that I_{C_4} is of the form $\tilde{M}$ for some graded R-module M, with $R = \mathbb{C}[x_0, x_1, x_2, x_3]$, see [EO25, Theorem 19.5]. We can therefore look at a free resolution of M to get a resolution of I_{C_4}. First, consider the map

$$R^{\oplus 2}(-2) \to M$$

given by the matrix
$$\begin{pmatrix} x_1x_3 & x_0x_2 \end{pmatrix}.$$

This map is surjective, as we see that the image contains the generators of M. Now, consider $(f, g) \in R^{\oplus 2}(-2)$, which maps to $x_0x_2f + x_1x_3g$. Then we know that at least $f = x_1x_3$ and $g = -x_0x_2$ maps to zero, and so we continue the resolution:

$$R(-4) \xrightarrow{\begin{pmatrix} x_1x_3 \\ -x_0x_2 \end{pmatrix}} R(-2)^{\oplus 2} \xrightarrow{\begin{pmatrix} x_1x_3 & x_0x_2 \end{pmatrix}} M \to 0.$$

Now we want to check if these are the only elements of the kernel. If $x_0x_2f + x_1x_3g = 0$, we have that $x_0x_2f = -x_1x_3g$. Since x_0, x_1, x_2, x_3 are all non-zero divisors in R, we must have that f is divisible by x_1x_3 and g is divisible by x_0x_2. Therefore, we can write $f = x_1x_3h$ and $g = -x_0x_2h$ for some $h \in R$. This means that the kernel of the map is generated only by the element we found, and so the resolution is exact. Then we have the following free resolution of M:

$$0 \to R(-4) \xrightarrow{\begin{pmatrix} x_1x_3 \\ -x_0x_2 \end{pmatrix}} R(-2)^{\oplus 2} \xrightarrow{\begin{pmatrix} x_1x_3 & x_0x_2 \end{pmatrix}} M \to 0.$$

Passing to the associated sheaves, we get the following resolution of the ideal sheaf of the 4-gon:

$$0 \to \mathcal{O}_{\mathbb{P}^3}(-4) \xrightarrow{\begin{pmatrix} x_1x_3 \\ -x_0x_2 \end{pmatrix}} \mathcal{O}_{\mathbb{P}^3}(-2)^{\oplus 2} \xrightarrow{\begin{pmatrix} x_1x_3 & x_0x_2 \end{pmatrix}} I_{C_4} \to 0.$$

Now we can twist this resolution by $\mathcal{O}_{\mathbb{P}^3}(d)$ to get

$$0 \to \mathcal{O}_{\mathbb{P}^3}(d-4) \xrightarrow{\begin{pmatrix} x_1x_3 \\ -x_0x_2 \end{pmatrix}} \mathcal{O}_{\mathbb{P}^3}(d-2)^{\oplus 2} \xrightarrow{\begin{pmatrix} x_1x_3 & x_0x_2 \end{pmatrix}} I_{C_4}(d) \to 0. \tag{4.2}$$

From this, we can get the long exact sequence in cohomology:

$$\begin{tikzcd}
0 \arrow[r] & H^0(\mathcal{O}_{\mathbb{P}^3}(d-4)) \arrow[r] & H^0(\mathcal{O}_{\mathbb{P}^3}(d-2))^{\oplus 2} \arrow[r] & H^0(I_{C_4}(d)) \arrow[dll, out=-10, in=170] \\
& H^1(\mathcal{O}_{\mathbb{P}^3}(d-4)) \arrow[r] & H^1(\mathcal{O}_{\mathbb{P}^3}(d-2))^{\oplus 2} \arrow[r] & H^1(I_{C_4}(d)) \arrow[dll, out=-10, in=170] \\
& H^2(\mathcal{O}_{\mathbb{P}^3}(d-4)) \arrow[r] & \dots
\end{tikzcd}$$

By [EO25, Theorem 20.23], we know that $H^i(\mathcal{O}_{\mathbb{P}^3}(d)) = 0$ for $i > 0$ and $d \geq (-3)$. Therefore, when $d - 4 \geq -3$, i.e. $d \geq 1$, we have that $H^1(I_{C_4}(d)) = 0$. Thus, we get the following proposition:

Proposition 4.3.9. *Let C_4 be a regular 4-gon in $\mathbb{P}^3$. Then for $d \geq 1$, we have that $h^1(\mathcal{I}_{C_4}(d)) = 0$, and we can find a degree d surface containing C_4 for $d \geq 2$.*

Proof. We have shown above that for $d \geq 1$, $H^1(I_{C_4}(d)) = 0$. Then the second part follows directly from Corollary 4.3.6 by the following observation: For $d = 1$, we have that $h^0(\mathcal{O}_{\mathbb{P}^3}(1)) = \binom{4}{3} - 4 = 0$, so there are no degree 1 surfaces (planes) containing a general 4-gon. $\square$

This is perhaps unsurprising, and in fact we could already show that a regular 4-gon can be included in a degree 2 surface, for example a smooth quadric surface:

Corollary 4.3.10. *Let C_4 be a regular 4-gon in $\mathbb{P}^3$. Then we can find a smooth degree 2 surface S_2 such that $C_4 \subseteq S_2$.*

Proof. Using the points from Lemma 4.3.8, we can include the 4-gon in $V(x_1x_3 - x_0x_2)$, and by generality of these points we can find such a smooth quadric surface for any general 4-gon. □

Still, there are interesting conclusions to be drawn from this. For example, we can find $h^0(\mathcal{O}_{\mathbb{P}^3}(d))$ for all $d \geq 1$, and therefore know the exact dimension of the space of degree d surfaces containing a regular 4-gon:

Corollary 4.3.11. *Let C_4 be a regular 4-gon in $\mathbb{P}^3$. Then for $d \geq 2$, we have that*

$$h^0(\mathcal{I}_{C_4}(d)) = \binom{d+3}{3} - 4d.$$

For example, for $d = 2$, we have that $h^0(\mathcal{I}_{C_4}(2)) = \binom{5}{3} - 8 = 10 - 8 = 2$, so the vector space of quadrics containing a regular 4-gon is 2-dimensional.

This section also reveals why we have not yet found a general result for $h^1(\mathcal{I}_{C_m})$, which is that we have so far relied on the generators of the ideal I_{C_4} to compute it. The generators of ideals are generally troublesome, and it is hard to prove general facts by using them. A concrete reason for why we had trouble generalizing in our problem, was that when checking the ideal sheaves of higher m in Macaulay2, we found that they do not grow linearly. We give a short table of the number of generators of the ideals of m-gons for small m:

Table 4.2: Number of generators of the ideals of m-gons for small m.

m	Number of generators
4	2
5	5
6	5
7	7
8	7
9	11
10	6
11	15

Note that these are the (minimal) number of all generators, not just for degree d.

4.4 Surfaces including m-chains

A 'happy accident' we made when studying m-gons in Macaulay2 was to forget to add the final line which then intersects the first line, closing the loop. Before realizing what exactly had happened, we noted that the results were strange and suspected an error in our code, which we then promptly fixed. Still, the results were a bit surprising, implying that for some chains we find only singular surfaces of certain degrees. After getting more expected results for m-gons, we decided to study these new formations further, resulting the theory in this section. We begin by defining m-chains.

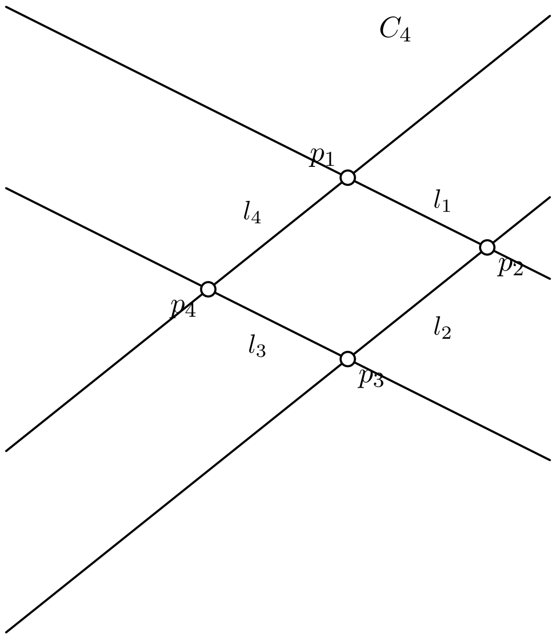


Figure 4.3: A 4-gon C_4 in $\mathbb{P}^3$, drawn on a plane.

Definition 4.4.1. An m-chain is a union of m lines $l_1, l_2, \ldots, l_m$ such that l_i intersects l_{i+1} for $1 \leq i < m$. We will write $Q_m = \bigcup_{i=1}^m l_i$ and denote the intersection points as $p_i = l_i \cap l_{i+1}$ for $1 \leq i < m$.

Definition 4.4.2. A *regular* m-chain is an m-chain such that no line l_i intersects any l_j for $|i-j| > 1$, and there are no double lines $l_i \neq l_j$ for $i \neq j$.

Definition 4.4.3. We define $\mathcal{Q}_m \subseteq \mathbb{G}(1,3)^m$ to be the subset of m-tuples of lines $(l_1, \ldots, l_m)$ such that they form an m-chain.

First we need the following lemma.

Lemma 4.4.4. *The variety $\Sigma_L = \{l \in \mathbb{G}(1,3) \mid l \cap L \neq \emptyset\}$ for a fixed line L is irreducible of dimension 3.*

Proof. First, consider the incidence correspondence $\Gamma = \{(l,p) \in \mathbb{G}(1,3) \times L \mid p \in l\}$. The projection map $g \colon \Gamma \to \mathbb{G}(1,3)$ given by $g(l,p) = l$ gives us that Σ_L is the image of g. Next, the fiber of the projection map $f \colon \Gamma \to L$ given by $f(l,p) = p$ is the set of lines through a point $p \in L$, meaning the fiber is isomorphic to $\mathbb{P}^2$. Since L is isomorphic to $\mathbb{P}^1$, it is irreducible of dimension 1. Since f is proper, and all the fibers are irreducible of dimension 2, we have from Lemma 1.3.6 that Γ is irreducible of dimension $2+1=3$. It follows that Σ_L is irreducible, and as a general fiber of g is a single point, it follows that $\dim \Sigma_L = \dim g(\Gamma) = \dim \Gamma = 3$. □

Remark 4.4.5. This variety Σ_L is a Schubert variety, often denoted σ_1 in the literature. It is part of the Schubert calculus on Grassmannians, which is a powerful tool for computing intersection numbers of subvarieties of Grassmannians. It is beyond the scope of this thesis, but refer the interested reader to [EH16, Section 3.3] for more information.

Lemma 4.4.6. *$\mathcal{Q}_m$ is an irreducible variety of dimension $3m+1$ for $m \geq 1$.*

Proof. We first observe that $\mathcal{Q}_1 = \mathbb{G}(1,3)$, which is irreducible of dimension 4. We then proceed by induction on m, assuming that $\mathcal{Q}_m$ is irreducible of dimension M for some integer M.

Let $m \geq 2$, and consider the projection map $\pi_m \colon \mathcal{Q}_m \to \mathcal{Q}_{m-1}$ given by $\pi(l_1, \dots, l_m) = (l_1, \dots, l_{m-1})$. The fiber over a point is given by $\pi^{-1}(l_1, \dots, l_{m-1}) = \{(l_1, \dots, l_{m-1}, l_m) \in \mathcal{Q}_m \mid l_m \cap l_{m-1} \neq \emptyset\}$ which is isomorphic to $\Sigma_{l_{m-1}} = \{l \in \mathbb{G}(1,3) \mid l \cap l_{m-1} \neq \emptyset\}$. From Lemma 4.4.4 we know that $\Sigma_{l_{m-1}}$ is irreducible of dimension 3, and by the induction hypothesis that $\mathcal{Q}_{m-1}$ is irreducible of dimension M. Then, since π_m is proper, with irreducible fiber of dimension 3, we again have that $\mathcal{Q}_m$ is irreducible of dimension $M+3$ by Lemma 1.3.6. $M = 4$ for $m = 1$, and so we get that $\dim \mathcal{Q}_m = 4 + 3(m-1) = 3m+1$. □

Corollary 4.4.7. *The set of regular m-chains in $\mathcal{Q}_m$ is a dense open subset of $\mathcal{Q}_m$.*

Proof. Let $R_m \subset \mathcal{Q}_m$ be the locus of regular m-chains. By definition a tuple $(l_1, \dots, l_m) \in \mathcal{Q}_m$ is regular if and only if: (i) $l_i \neq l_j$ for all $i \neq j$; (ii) $l_i \cap l_j = \emptyset$ whenever $j \not\equiv i \pm 1$ (i.e. only adjacent pairs meet).

Condition (i) is can be described by the locus of a finite number of diagonals $\Delta_{ij} = \{(l_1, \dots, l_m) \mid l_i = l_j\}$ in $\mathbb{G}(1,3)^m$, and so its complement is open. Condition (ii) is described by the complements of a finite number of zero loci of the equations $l_i \cap l_j \neq \emptyset$ for adjacent pairs (i,j), and so this condition is also open. □

As discussed above, we note that an m-chain is an $(m+1)$-gon excluding the line l_m intersecting l_1. One can then modify the code to create m-chains instead of $(m+1)$-gons, which is included in Section A.2. The results from this are similar to those of m-gons, in that for higher degrees, we find smooth surfaces, up until we find no surfaces. There are however interesting exceptions for degree 2 and 3 surfaces. Here we find singular surfaces when considering 4-chains and 6-chains respectively. We will handle the cases separately. First, we show a similar proposition to that of m-gons in Proposition 4.3.4.

First we add the promised corollary to Lemma 4.3.3:

Corollary 4.4.8. *The global sections of the structure sheaf of a regular m-chain Q_m has dimension $h^0(Q_m, \mathcal{O}_{Q_m}(d)) = md - 1$, for $d \geq 1$.*

Proof. The proof is similar to that of Lemma 4.3.3, only now we have the short exact sequence

$$0 \to \mathcal{O}_{Q_m}(d) \to \bigoplus_{i=1}^{m} \mathcal{O}_{l_i}(d) \to \bigoplus_{i=1}^{m-1} \mathcal{O}_{p_i} \to 0,$$

since we only have $m-1$ intersection points in an m-chain. Following the same argument as in the proof of Lemma 4.3.3, we get that $h^0(Q_m, \mathcal{O}_{Q_m}(d)) = h^0(\bigoplus_{i=1}^m \mathcal{O}_{l_i}(d)) - h^0(\bigoplus_{i=1}^{m-1} \mathcal{O}_{p_i}) = md - 1$. □

Proposition 4.4.9. *Let Q_m be a regular m-chain in $\mathbb{P}^3$. Then we can find a hypersurface S_d of degree d such that $Q_m \subseteq S_d$ when $m < (\binom{d+3}{3} - 1) \cdot \frac{1}{d}$, or equivalently $m < \frac{(d+3)(d+2)(d+1)-1}{6d}$.*

Proof. The proof is very similar to that of Proposition 4.3.4, so we will be brief here. Again, by abuse of notation, we write $\mathcal{O}_{Q_m}$ for $i_*\mathcal{O}_{Q_m}$, where $i : Q_m \to \mathbb{P}^3$ is the inclusion. We have the short exact sequence

$$0 \to \mathcal{I}_{Q_m}(d) \to \mathcal{O}_{\mathbb{P}^3}(d) \to \mathcal{O}_{Q_m}(d) \to 0,$$

Table 4.3: Results from Macaulay2 over $\mathbb{Q}$ testing existence and smoothness of surfaces containing an m-gon. Each row corresponds to one experiment: degree d, number of points used, whether the chosen form was singular, whether the space of degree-d forms in the ideal was zero.

d	m	Singular	Zero
2	4	true	false
2	5	false	true
3	6	true	false
3	7	false	true
4	8	false	false
4	9	false	true
5	11	false	false
5	12	false	true
6	13	false	false
6	14	false	true

which by the same analysis as before gives us

$$h^0(\mathbb{P}^3, \mathcal{I}_{Q_m}(d)) = h^0(\mathbb{P}^3, \mathcal{O}_{\mathbb{P}^3}(d)) + h^1(\mathbb{P}^3, \mathcal{I}_{Q_m}(d)) - h^0(Q_m, \mathcal{O}_{Q_m}(d)).$$

From Corollary 4.4.8 we have that $h^0(Q_m, \mathcal{O}_{Q_m}(d)) = md - 1$. Again, by [EO25, Corollary 20.26] we know $h^0(\mathbb{P}^3, \mathcal{O}_{\mathbb{P}^3}(d)) = \binom{d+3}{3}$. Thus, we have

$$h^0(\mathbb{P}^3, \mathcal{I}_{Q_m}(d)) \geq \binom{d+3}{3} - md - 1 + h^1(\mathbb{P}^3, \mathcal{I}_{Q_m}(d)).$$

We still know that $h^1(\mathbb{P}^3, \mathcal{I}_{Q_m}(d)) \geq 0$, so

$$h^0(\mathbb{P}^3, \mathcal{I}_{Q_m}(d)) = \binom{d+3}{3} - md - 1,$$

and we can guarantee a surface when $m < (\binom{d+3}{3} - 1) \cdot \frac{1}{d}$. $\square$

4.4.1 Computational results

We ran similar Macaulay2 code as for m-gons, modified to create m-chains instead. The code is included in Section A.2. The results are summarized in the table below.

When analyzing the results, we see that for 4-chains in degree 2 surfaces, and 6-chains in degree 3 surfaces, we only find singular surfaces including the chains. For higher degrees, we find smooth surfaces including the chains, up until we find no surfaces including the chains. We checked this as follows: We first run the regular script for m-chains over $\mathbb{Q}$, as can be found in Section A.2. When running Macaulay2 on a script, all values are saved until it is changed or the program is closed. So we checked our singular surfaces by using the command `factor(f)`, which factors the polynomial f defining the surface. We found that the polynomials factored into linear terms, respectively 2 and 3 such terms, meaning that the surfaces were unions of 2 and 3 planes. Next, we applied the command `mingens Iz` to find the minimal generators of the ideal I_Z of the chain Z. We found that in both cases, there was only one generator in degree 2 and 3 respectively, meaning that there was a unique quadric and cubic surface including the 4-chain and 6-chain respectively. We gather these into the following lemmas.

Lemma 4.4.10. *For some 4-chain in $\mathbb{P}^3_{\mathbb{Q}}$, there is a unique singular quadric surface $S_2 \subset \mathbb{P}^3$ such that it includes the 4-chain.*

Lemma 4.4.11. *For some 6-chain in $\mathbb{P}^3_{\mathbb{Q}}$, there is a unique singular cubic surface $S_3 \subset \mathbb{P}^3$ such that it includes the 6-chain.*

4.4.2 The case of 4-chains

Proposition 4.4.12. *For a regular 4-chain in $\mathbb{P}^3$, we cannot find a smooth quadric surface $S_2 \subset \mathbb{P}^3$ such that it includes the 4-chain.*

Proof. First, assume that S_2 is smooth. Since S_2 is a smooth quadric surface in $\mathbb{P}^3$, it is of the form $S_2 = \mathbb{P}^1 \times \mathbb{P}^1$. This means that S_2 has two rulings, meaning that there are two families of lines on S_2, such that any two lines in the same family do not intersect, and any two lines in different families intersect at exactly one point. Denote the two families as F_1 and F_2. Then we can see that a 4-chain must consist of two lines from F_1 and two lines from F_2, since any two lines in the same family do not intersect. This means that the 4-chain must be of the form l_1, l_2, l_3, l_4 where $l_1, l_3 \in F_1$ and $l_2, l_4 \in F_2$. However, this means that l_4 intersects l_1, since they are in different families, meaning that the 4-chain is in fact a 4-gon. This is a contradiction, since this is not a regular 4-chain. Thus, we cannot find a smooth quadric surface $S_2 \subset \mathbb{P}^3$ such that it includes the 4-chain. □

This contrasts the case of 4-gons, where we could find smooth quadric surfaces including the 4-gon, as shown in Corollary 4.3.10 in Section 4.3.2.

Proposition 4.4.13. *For a 4-chain in $\mathbb{P}^3$, we can find a singular quadric surface $S_2 \subset \mathbb{P}^3$ such that it includes the 4-chain.*

Proof. We pair the lines of the 4-chain as (l_1, l_2) and (l_3, l_4). First we assume that the 4-chain is regular. Each pair of lines defines a plane in $\mathbb{P}^3$, denoted as $H_1 = \mathrm{span}(l_1, l_2)$ and $H_2 = \mathrm{span}(l_3, l_4)$. Since the 4-chain is regular, we have that $H_1 \neq H_2$. Then the union of the two planes $S_2 = H_1 \cup H_2$ is a singular quadric surface in $\mathbb{P}^3$ that includes the 4-chain.

Now we consider a non-regular 4-chain. If the lines are not distinct, we choose H_1 and H_2 to be any planes including the pairs of lines respectively. If the lines intersect more than the adjacent lines, the plane might be the same, in that case we choose H_2 to be any plane different from H_1. □

4.4.3 The case of 6-chains

We now consider the incidence variety of cubic surfaces including a 6-chain. Cubic surface in $\mathbb{P}^3$ are parametrized by the projective space $\mathbb{P}(\mathbb{C}[x_0, \dots, x_3]_3) \cong \mathbb{P}^{19}$. Let $\mathcal{N} = \{(X, Q_6) \in \mathbb{P}^{19} \times \mathcal{Q}_6 \mid Q_6 \subset X\}$ be the incidence variety of cubic surfaces including a 6-chain. Note that this is equivalent to the set $\{(X, l_1, \dots, l_6) \in \mathbb{P}^{19} \times \mathbb{G}(1,3)^6 \mid \bigcup_{i=1}^6 l_i$ is a 6-chain included in $X\}$, it is just notationally simpler to use the first definition. We have the following diagram:

Now we will examine 6-chains in cubic surfaces. We will show that a regular 6-chain is not included in a smooth cubic surface in two ways. The first result uses the incidence variety, and the other shows that a 6-chain in a smooth cubic surface cannot be regular. We will first show that we can find singular cubic surfaces including any 6-chain, which

$$\begin{tikzcd} & \mathcal{N} \arrow[dl, "p"'] \arrow[dr, "q"] & \\ \mathbb{P}^{19} & & \mathcal{Q}_6 \end{tikzcd}$$

Figure 4.4: Incidence diagram with the morphisms $p : \mathcal{N} \to \mathbb{P}^{19}$ and $q : \mathcal{N} \to \mathcal{Q}_6$.

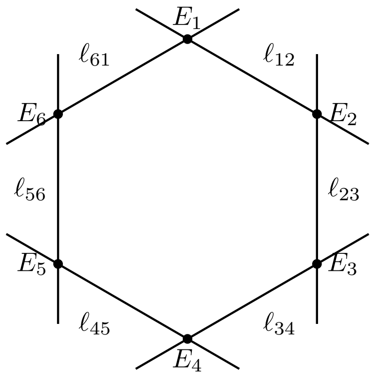


Figure 4.5: Six exceptional lines $E_1, \ldots, E_6$, and the strict transform of six lines, used to form a 12-gon.

will be useful to prove that we cannot find a smooth cubic surface including a regular 6-chain.

Lemma 4.4.14. *Any 6-chain is included in a union of 3 planes in* $\mathbb{P}^3$.

Proof. As in Proposition 4.4.13 We pair the lines of the 6-chain as (l_1, l_2), (l_3, l_4) and (l_5, l_6). Each pair of lines define a plane in $\mathbb{P}^3$. Then the union of the three planes is a singular cubic surface in $\mathbb{P}^3$ that includes the 6-chain. □

Another lemma we will need is to show that we can find a 6-chain in any smooth cubic surface. We show this by finding a 12 chain. This will be quite degenerate, but we are not looking for a regular chain here.

Lemma 4.4.15. *For a smooth cubic surface in* $\mathbb{P}^3$*, we can find a 12-chain such that it is included in the surface.*

Proof. We will build this 12-chain explicitly. Recall from Proposition 3.4.2 that a smooth cubic surface is isomorphic to the blowup of $\mathbb{P}^2$ at six points, and that there are 27 lines on the surface, as described in Proposition 3.4.3. We will use the notation from these propositions. Consider the exceptional curves $E_1, E_2, E_3, E_4, E_5, E_6$, and the strict transforms of the lines $\ell_{12}, \ell_{23}, \ell_{34}, \ell_{45}, \ell_{56}$ through pairs of points. Then one forms a 12-chain by considering the lines in the following order:

$$E_1, \ell_{12}, E_2, \ell_{23}, E_3, \ell_{34}, E_4, \ell_{45}, E_5, \ell_{56}, E_6, C_1, \ell_{16}.$$

We include a figure of this in Fig. 4.5 for clarity. □

Note that one could continue building the chain further, but we stop at 12 for simplicity, and especially for the nicer picture in Fig. 4.5. Note also that the 12-chain is also a 12-gon, since the last line intersects the first line.

Lemma 4.4.16. *If a 6-chain is included in a smooth cubic surface, it is included in an (at least) 1-dimensional family of smooth cubic surfaces.*

Proof. Let Q_6 be a 6-chain in a smooth cubic surface X. From Lemma 4.4.14, we know that it is also included in a singular surface Y. Then let $\mathcal{D}$ be the pencil $\{aX + bY\}$ for $a, b \in \mathbb{C}$. All the surfaces in this linear system include our 6-chain, by the observation that if $X = V(f)$ and $Y = V(g)$, then inclusion is equivalent to $f|_{Q_6} = g|_{Q_6} = 0$, and $af|_{Q_6} + bg|_{Q_6} = 0$. Then we use that smoothness is an open property to see that most of the surfaces are smooth. This gives us a 1-dimensional family of smooth surfaces including our 6-chain. $\square$

Proposition 4.4.17. *For a regular 6-chain in $\mathbb{P}^3$, we cannot find a smooth cubic surface $S_3 \subset \mathbb{P}^3$ such that it includes the 6-chain.*

Proof. First, consider the diagram in Fig. 4.4. The map $p\colon \mathcal{N} \to \mathbb{P}^{19}$ is given by $p(X, Q_6) = X$. Then $p^{-1}(\mathcal{M})$ for the smooth cubic surfaces $\mathcal{M} \subset \mathbb{P}^{19}$ is open in $\mathcal{N}$, we denote $p^{-1}(\mathcal{M}) = \mathcal{N}_{\mathcal{M}}$. The fiber over a point $X \in \mathcal{M}$ is given by the set of 6-chains included in X. Since a smooth cubic surface includes exactly 27 lines, there is only finitely many ways to choose 6 lines from these 27 to form a 6-chain. Thus, the dimension of the fiber $\dim p^{-1}(X)$ is 0. Then $\mathcal{N}_{\mathcal{M}}$ has dimension 19.

We know that $\dim \mathcal{Q}_6 = 19$ from Lemma 4.4.6, and we want to show that $q(\mathcal{N}_{\mathcal{M}})$, which is the set of 6-chains included in smooth cubic surfaces, is not dense in $\mathcal{Q}_6$. We know from Lemma 4.4.16 that an element $Q_6 \in q(\mathcal{N}_{\mathcal{M}})$, i.e. a 6-chain included in a smooth cubic surface, is included in at least a 1-dimensional family of smooth cubic surfaces. Thus, the fiber $q^{-1}(Q_6)$ has dimension at least 1. Then $q(\mathcal{N}_{\mathcal{M}})$ must be of dimension at most 18. Since the subset of regular m-gons is an open subset of $\mathcal{Q}_m$, and $\mathcal{Q}_m$ is irreducible, the subset of regular 6-chains also has dimension 19. Therefore, we cannot find a smooth cubic surface including a regular 6-chain in $\mathbb{P}^3$. $\square$

Remark 4.4.18. We use the set of m-chains, not regular m-chains, so that the fiber of p is non-empty.

Note that this means that the 6-chains which *are* included in smooth cubic surfaces, are not general. To see why this might be, note that as we saw in the proof of **??**, requiring that pairs of lines in $\mathbb{P}^3$ do not meet is a Zariski-open condition. This is exactly the separating factor of the 6-chains included in smooth cubic surfaces. The requirement of non-intersection is also included in the definition of a regular m-chain.

The fact that we cannot find a surface including a general 6-chain may not be very intuitive, but the following lemma gives a good reason.

Lemma 4.4.19. *For a smooth cubic surface, we can find at most a regular 5-chain.*

Proof. First we show that we can find a regular 5-chain. Again we consult Fig. 3.1 which shows a smooth surface as the blow-up $\mathbb{P}^2$ in six points. Then the chain $E_1 \cup \ell_{12} \cup E_2 \cup \ell_{23} \cup E_3$ meets the criteria of a regular 5-chain.

Now we want to show that we cannot find a regular 6-chain. First, denote the 27 lines of the cubic surface as $C_1, \dots, C_{27}$. Let Q_5 be a regular 5-chain, consisting of $C_1, \dots, C_5$. Look at the class $D = C_1 + C_2 - C_4 - C_5$. Then $D^2 = 0$, and $HD = 0$ for $H = f^*(\mathcal{O}_{\mathbb{P}^3}(1))$. Since H is ample, the Hodge index theorem, Theorem 3.1.9, tells us that $D = 0$, and we have the equality $C_1 + C_2 = C_4 + C_5$. Let $C_i \neq C_4$ be another line which intersects C_5. Then it must also intersect either C_1 or C_2 by the equality, and thus cannot be used to make a regular 6-chain. $\square$

One could also use this lemma to prove Proposition 4.4.17 for regular 6-chains, similarly to our proof that a regular 4-chain is not included in a smooth quadric surface. i.e. we note that if we have a 6-chain in a smooth cubic surface, the lemma above tells us that this cannot be a regular 6-chain.

Our testing in Macaulay2 also suggested that the singular surface was unique, which we will prove in the following proposition.

Proposition 4.4.20. *A regular 6-chain in $\mathbb{P}^3$ is only included in a unique surface.*

Proof. Consider the diagram in Fig. 4.4. The map q has fiber $q^{-1}(Q_6) = \{X \mid Q_6 \in X\}$ for $Q_6 \in \mathcal{Q}_6$. q is a projective morphism, since it is the composition of the closed embedding $i\colon \mathcal{N} \to \mathbb{P}^N \times \mathcal{Q}_6$, and the projective map $\pi\colon \mathbb{P}^N \times \mathcal{Q}_6 \to \mathcal{Q}_6$. A projective morphism is proper by [Har77, Theorem II.4.9]. Then taking the dimension of the fiber of q is upper semi-continuous.

From Lemma 4.4.11, we have an example where the fiber is a single point. Call the relevant 6-chain Z. Though this computation was done over $\mathbb{Q}$, the result extends to $\mathbb{C}$. To see why, first note that when we find that there is only one unique surface including Z, it is equivalent to saying that the kernel of the map $\phi\colon H^0(\mathbb{P}^3, \mathcal{O}(3)) \to H^0(Z, \mathcal{O}(3)|_Z$ is 1-dimensional. Note also that the fiber of $q(Z)$ is the set $\{f \in \mathbb{C}[x_0, x_1, x_2, x_3]_3 \mid f|_Z = 0\}$, which are the polynomials which are zero on Z. The equations requiring $f|_Z = 0$ have rational coefficients, since our Z is defined over the rationals. Therefore, the kernel of ϕ is also 1-dimensional over $\mathbb{C}$. Because of this, there exists at least one 6-chain with a single point as fiber, and so the fiber over this point is 0-dimensional.

By Lemma 4.4.14, we know that any 6-chain is included in at least one cubic surface, so the fiber of q is non-empty for all points in $\mathcal{Q}_6$. Then, by upper semi-continuity of fiber dimension, Theorem 1.3.7, we have that since we know one fiber, $q^{-1}(Z)$ is 0-dimensional, there exists an open neighborhood U of Z, such that $\dim q^{-1}(Z') \leq 0$ for all $Z' \in U$. Since every fiber is non-empty, all of these fibers are 0-dimensional, and so there are a finite number of surfaces including a regular Q_6. Then we note that the fiber of q is the set of solutions of linear equations, and so it defines a projective space. Since the fiber is 0-dimensional, it must be the point $\mathbb{P}^0$, and so there is only one surface. $\square$

Chapter 5

Conclusion

Conjecture 5.0.1. *Let C_m be a regular m-gon in $\mathbb{P}^3$, then we have*

$$h^1(\mathbb{P}^3, \mathcal{I}_{C_m}(d)) = \max\left(0, \binom{d+3}{3} - md\right).$$

This would mean that the bound we have found for the existence of surfaces containing the m-gon is sharp, and we would get the corollary:

Corollary 5.0.2. *For a regular m-gon C_m in $\mathbb{P}^3$, we can not find any degree d surface S_d such that $C_m \subseteq S_d$ when $m \geq \binom{d+3}{3} \cdot \frac{1}{d}$.*

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Appendix A

Computational methods and results

This appendix documents the Macaulay2 scripts used for the computational experiments, along with a selection of results. One can find the full code repository at https://github.com/bilgetit/MasterOppgave/Progging.

A.1 The m-gon script

Listing A.1: Script generating an m-gon and testing degree-d forms

```
-- Step 1: Define the ring and parameters
kk = QQ;
RingP3 = kk[x_0..x_3];
d = 2;  -- Degree of the hypersurface
nrPoints = 12;

-- Step 2 and 3: Generate random points and create pairs of
    subsequent points
-- Function for generating i random points in P^3
randomPoints = i -> (
     v := for j in 0..3 list random(kk);
     -- Ensure the point is not zero
     while all(v, x -> x == 0) do (
          v = for j in 0..3 list random(kk)
     );
     v
)
-- Pick nrPoints random points in P^3
myPoints = apply(nrPoints, randomPoints);
-- Create pairs of subsequent points, and also the pair consisting
     of the last and first point
subsequentPairs = for i from 0 to (#myPoints - 2) list {myPoints#i
    , myPoints#(i+1)};
-- Add final pair
subsequentPairs = append(subsequentPairs, {myPoints#0, myPoints#(#
    myPoints-1)});

-- Step 3: Generate matrices from point pairs
-- Function to convert two points to a matrix with first row {x_0
    .. x_3}
pointsToMatrix = points -> matrix {{x_0 .. x_3}, points#0, points
    #1};
```

```
-- Convert each pair of points to a matrix
pointsMatrix = apply(subsequentPairs, pointsToMatrix);

-- Step 4: Get the ideals of the lines
-- Given a matrix with first row {x_0 .. x_3} and next two rows
   points, return the ideal of the line through the points
lineFromPoints = pointsMatrix -> minors(3, pointsMatrix);
allLines = apply(pointsMatrix, lineFromPoints);

-- Step 5: Create the ideal of the union of all lines
-- Intersect the ideals of the lines to get ideal of the union of
   all lines
Iz = intersect allLines;
IzGens = mingens Iz; -- If one wants to see the number of
   generators
-- hf = hilbertFunction(d, RingP3/Iz);
-- totalForms = binomial(d+3,3);
-- numInIz = totalForms - hf;

-- Step 6: Pick a degree d element in Iz, which defines a surface
   containing all the lines, and check properties
-- Pick a random degree d element in Iz
f = random(d, Iz);
-- Create the variety defined by f
Vf = variety ideal f;

-- Check if the surface is smooth and not the zero polynomial
codimension = codim singularLocus ideal f;
isZero = (f == 0)
isSingular = (codimension < 4)
isSmooth(Vf) and not isZero

IzGens
-- numInIz
```

A.1.1 Tables of results

Table A.1: Results over $\mathbb{Q}$ for m-gons.

oprule d	m	Singular	Zero
2	4	false	false
2	5	false	true
3	6	false	false
3	7	false	true
4	8	false	false
4	9	false	true
5	11	false	false
5	12	false	true
6	13	false	false
6	14	false	true

Table A.2: Results over $\mathbb{Z}_{32749}$ for m-gons (same columns).

oprule d	m	Singular	Zero
2	4	false	false
2	5	false	true
3	6	false	false
3	7	false	true
4	8	false	false
4	9	false	true
5	11	false	false
5	12	false	true
6	13	false	false
6	14	false	true
7	17	false	false
7	18	false	true
8	20	false	false
8	21	false	true
9	24	false	false
9	25	false	true
10	28	false	false
10	29	false	true
11	33	false	false
11	34	false	true
12	37	false	false
12	38	false	true
13	43	false	false
13	44	false	true
14	48	false	false
14	49	false	true
15	54	false	false
15	55	false	true

A.2 The m-chain script

Listing A.2: Script for an m-chain (number of lines $= m - 1$)

```
-- Step 1: Define the ring and parameters
kk = QQ;
-- kk = ZZ/32749;
RingP3 = kk[x_0..x_3];
d = 3;  -- Degree of the hypersurface
nrLines = 6;
nrPoints = nrLines+1;
maxNoPoints = (binomial(d+3, 3) - 1)/d;

-- Step 2 and 3: Generate random points and create pairs of subsequent points
-- Function for generating i random points in P^3
randomPoints = i -> (
    v := for j in 0..3 list random(kk);
    -- Ensure the point is not zero
    while all(v, x -> x == 0) do (
        v = for j in 0..3 list random(kk)
    );
    v
)
-- Pick nrPoints random points in P^3
myPoints = apply(nrPoints, randomPoints);
-- Create pairs of subsequent points, and also the pair consisting of the last and first point
subsequentPairs = for i from 0 to (#myPoints - 2) list {myPoints#i, myPoints#(i+1)};

-- Step 3: Generate matrices from point pairs
-- Function to convert two points to a matrix with first row {x_0 .. x_3}
pointsToMatrix = points -> matrix {{x_0 .. x_3}, points#0, points#1};
-- Convert each pair of points to a matrix
pointsMatrix = apply(subsequentPairs, pointsToMatrix);

-- Step 4: Get the ideals of the lines
-- Given a matrix with first row {x_0 .. x_3} and next two rows points, return the ideal of the line through the points
lineFromPoints = pointsMatrix -> minors(3, pointsMatrix);
allLines = apply(pointsMatrix, lineFromPoints);

-- Step 5: Create the ideal of the union of all lines
-- Intersect the ideals of the lines to get ideal of the union of all lines
Iz = intersect allLines;

-- Step 6: Pick a degree d element in Iz, which defines a surface containing all the lines, and check properties
-- Pick a random degree d element in Iz
f = random(d, Iz);
-- Create the variety defined by f
Vf = variety ideal f;
```

```

-- Check if the surface is smooth and not the zero polynomial
codimension = codim singularLocus ideal f;
isZero = (f == 0)
isSingular = (codimension < 4)
isSmooth(Vf) and not isZero

F = sheaf(Iz);
H0 = HH^0(F(d))
H1 = HH^1(F(d))

-- primaryDecomposition(Iz)
-- factor(f)
maxNoPoints = numeric((binomial(d+3, 3) - 1)/d)
```